University of Glasgow

School of Social and Political Sciences

Urban Studies

Algorithmic Governance, Urban Space, and Domicide:

A Systematic Literature Review of the Dual-Use of Urban Algorithmic Systems, 1990–2025

Elias Vasiloudes Nikolaides

Student ID Number (GUID): 3069953V

Presented in partial fulfilment of the requirements for the degree of MSc

Public Policy & Management

Supervised by Craig Gurney

Word Count: 14,756 (main body)

August 2026

Abstract

This dissertation investigates the structural dual-use of algorithmic urban governance technologies: the capacity of a single data infrastructure to serve civilian urban management and military spatial destruction alike, with no new procurement — only a change of inputs. Rather than arguing from selected cases, it maps how the literature of 1990–2025 has — and has not — connected the algorithmic systems that optimise cities with those that destroy them. A theoretical framework assembles five elements: the economics of information goods, societies of control, the machinic city, epistemic authority, and the urbicide canon. Following a modified PRISMA protocol, 874 database records and 1,839 citation-chained candidates were screened, yielding 275 included records. Coding across four axes reveals a literature whose elements are documented but segregated: the governance and destruction literatures run in parallel, with only five records addressing algorithmically mediated urbicide. The central empirical contribution is a map of a junction the literature has not yet built. Most consequentially, 'dual-use' itself operates both as an analytic concept and as a juridical warrant for striking civilian infrastructure. The dissertation recasts the datafied city as a latent targeting substrate and identifies procurement and oversight — not technology design — as the critical policy lever.

List of Figures

2.1 The five-element framework and the dual-use claim. Arrow's economics of information goods, Deleuze's societies of control, DeLanda's machinic city, Michael's epistemic authority, and the urbicide canon (King, Graham, Weizman) converge on a single data substrate — one that can be parameterised for welfare or warfare with no change to the infrastructure itself. 15

3.1 Publication-year distribution of the included corpus (n = 271 in-window records), stacked by provenance. The concentration of records after 2015 — driven by the smart-city wave on the governance side and by post-2021 work on the destruction side — reflects the technological epochs the window was chosen to span; four in-press records dated 2026 fall outside the identification window. 20

3.2 Publication volume for five key terms, 1990–2025, log scale (OpenAlex works matching each term in title or abstract; retrieved 6 August 2026). The governance terms ("smart cities", "algorithmic governance", "predictive policing") exhibit the post-2010 spike the window was designed to capture; the destruction terms ("urbicide", "domicide") remain two orders of magnitude smaller, foreshadowing the asymmetry documented in Chapter 4. 21

3.3 PRISMA 2020 flow diagram for the modified protocol. Final counts as of 28 July 2026: 874 unique records screened from database searches and 2,081 deduplicated citation-chaining candidates, yielding 275 included studies (270 coded). 29

4.1 Theme score distributions by corpus (share of coded records) 32

5.1 Mean coding scores by search strategy across the framework's four themes. The governance literature (Search A) saturates Themes 1–3 but is nearly silent on spatial destruction; the destruction literature (Search B) shows the inverse pattern; citation chaining deepens both sides without bridging them. 47

6.1 The systematic map. Two well-documented literatures run in parallel — algorithmic urban governance (Search A) and spatial destruction (Search B) — with the junction of algorithmically mediated urbicide substantively addressed by only five records of the coded 270, all but one published since 2021. 53

List of Tables

4.1 Mean theme scores by corpus (0–3 scale; n scoring $\geq$2 in parentheses) . . . 31

List of Terms and Abbreviations

Terms are defined as used in this dissertation; section references point to fuller discussion.

Black box
: A system whose inputs and outputs are visible but whose internal reasoning is not (Pasquale, 2015). Chapter 4 shows such opacity to be manufactured by procurement and statute, not inherited from the technology itself.

Citation chaining (snowballing)
: A search method that follows the literature's own citation web rather than keywords: *forward* chaining harvests every later work citing an anchor text; *backward* chaining harvests every reference the anchor lists (§3.3).

Counter-forensics
: The practice of turning state sensing and mapping infrastructures toward accountability — using the same data that targeting uses to document destruction instead (Weizman, 2017).

Dataism
: The belief in the objective quantification and tracking of all human behaviour as a source of truthful, neutral knowledge (Dijck, 2014).

Dividual
: Deleuze's term for the person decomposed into data points — samples, markets, banks — rather than addressed as an enclosed individual (Deleuze, 1990).

Domicide
: The deliberate destruction of home, and of the conditions that make dwelling possible (King, 2004).

Dual-use
: Used here in a strong sense: a structural property of information goods, whereby transfer between civilian and military application requires only a change of inputs (§2.2). The term also names the legal category under which strikes on civilian infrastructure are licensed — a duality the dissertation shows to be consequential (§5.3).

Epistemic authority
: Michael's term for the institutionally recognised right to define what counts as true knowledge about a conflict — claimed, in his account, by the military over the civilian political echelon (Michael, 2007).

Information good
: Arrow's economic category: non-rivalrous, costly to produce, near-costless to reproduce — the properties from which this dissertation derives structural dual-use (Arrow, 1962).

Kill chain
: The military sequence from target identification through tracking to strike; Chapter 4 shows it beginning upstream, in the categories that make targets thinkable.

Machinic city
: DeLanda's account of the city as an assemblage of material and informational flows that can be modelled, and therefore programmed, at population scale (DeLanda, 1992; DeLanda, 2014).

Oversight gap
: This dissertation's term (§1.1) for the asymmetry between the state's rapidly expanding algorithmic capacity to act upon citizens and the capacity of democratic institutions to understand, audit, and contest those actions.

Pattern of life
: Intelligence-community term for the recorded regularities of a person's movements and associations, against which algorithms flag the abnormal.

Societies of control
: Deleuze's successor to disciplinary society: power operating continuously, through codes and modulation, rather than through enclosed institutions (Deleuze, 1990).

Transduction
: The translation of the physical city and its populace into programmable code (§1.1).

Urbicide
: The killing of the city as such: of cosmopolitan mixing, of the means of modern urban living, and of those cast as unmodern (Graham, 2012).

Abbreviations

AI Artificial intelligence

EOS
: Estimation of the Situation (the IDF's staff knowledge-production process; §2.5)

FPV First-person view (a class of short-range attack drone)

GIS Geographic information system

IDF Israel Defense Forces

IHL International humanitarian law

IoT Internet of Things

LLM
: Large language model

PRISMA
: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (Page et al., 2021)

SLR Systematic literature review

Contents

Abstract **ii**

List of Terms and Abbreviations **vi**

1 Introduction: The Algorithm and the City **1**

1.1 Smart Urbanism and the Oversight Gap 1

1.2 The Dual-Use Problem 2

1.3 Research Questions and Structure 4

2 Theoretical Framework **7**

2.1 Assembling the Framework 7

2.2 Arrow: Information Goods and Structural Dual-Use 8

2.3 Deleuze: From Discipline to Control 9

2.4 DeLanda: Machinic Assemblages and the Targetable City 10

2.5 Michael: Epistemic Authority Through Data 11

2.6 King, Graham, and Weizman: Domicide, Urbicide, Vertical Sovereignty . . 13

2.7 Synthesis: The Analytical Lens 14

3 Methodology: A Systematic Mapping of the Literature on Algorithmisation, 1990–2025 **17**

3.1 Review Design 17

3.2 Conceptual Scope and Research Questions 18

3.3 Search Strategy 19

3.3.1 Databases and date range 19

3.3.2 Dual-search design 21
3.3.3 Calibration 22
3.3.4 Citation chaining 22
3.4 Screening and Selection 24
3.4.1 Deduplication and corpus assembly 24
3.4.2 Inclusion and exclusion criteria 24
3.4.3 Single-reviewer screening with AI assistance 25
3.4.4 Screening results 25
3.5 Full-Text Retrieval and Thematic Coding 26
3.6 Synthesis Approach and Limitations 27

4 Findings: The Systematic Map **30**
4.1 The Shape of the Corpus 30
4.2 Theme 1 — The Dual-Use Mechanism ($n = 86$ scoring $\geq$2) 32
4.2.1 Dual-use as design goal: the practitioner literature 32
4.2.2 The boomerang: war-zone technology comes home 33
4.2.3 Commercial data, state security 34
4.2.4 Dual-use as geostrategy 34
4.2.5 The hole: nobody prices the transfer 34
4.3 Theme 2 — Epistemic Authority and the Black Box ($n = 117$ scoring $\geq$2) . 35
4.3.1 The pattern is political 35
4.3.2 Opacity is manufactured, not inherited 36
4.3.3 Race is constitutive, not contaminating 36
4.4 Theme 3 — The City as Programmable System ($n = 102$ scoring $\geq$2) . . . 37
4.5 Theme 4 — Spatial Destruction and Home Unmaking ($n = 76$ scoring $\geq$2) 38
4.5.1 The urbicide canon and its paradigmatic case 38
4.5.2 Destruction as cycle: razing, renewal, razing 39
4.5.3 The warrant: dual-use inside the destruction 40
4.5.4 The thin edge: algorithmic destruction 40
4.6 The Map as Argument 42

5 Discussion **44**

5.1 The Claim, Tested . . . 44

5.1.1 Arrow: present in the phenomenon, absent in the explanation . . . 44

5.1.2 Deleuze and DeLanda: confirmed by convergence . . . 45

5.1.3 Michael: confirmed and extended . . . 45

5.1.4 King, Graham, Weizman: the canon holds, the link is missing . . . 46

5.2 The Empty Intersection . . . 46

5.3 Dual-Use as Warrant . . . 48

5.4 Policy Implications . . . 49

5.5 What the Map Cannot Say . . . 51

6 Conclusion and Research Agenda **52**

6.1 What the Dissertation Set Out to Do . . . 52

6.2 What It Found . . . 52

6.3 An Agenda for Research . . . 54

6.4 Closing . . . 55

Bibliography **56**

A AI-Use Statement **66**

B Systematic Search Strategy **67**

B.1 Search A — Civil-Military Urban Technology Nexus . . . 67

B.1.1 Web of Science (Core Collection) . . . 67

B.1.2 Scopus . . . 67

B.1.3 Policy Commons . . . 68

B.2 Search B — Spatial Destruction and the Datafied City . . . 68

B.2.1 Web of Science (Core Collection) . . . 68

B.2.2 Scopus . . . 68

B.2.3 Policy Commons . . . 69

B.3 Notes . . . 69

Chapter 1

Introduction: The Algorithm and the City

1.1 Smart Urbanism and the Oversight Gap

The contemporary urban landscape is increasingly defined by an imperative toward 'smartness' (Cugurullo et al., 2024). Propelled by a confluence of digital technologies — spatiotemporal tracking, the Internet of Things (IoT), and Big Data analytics — city planners and policymakers have embraced algorithmic solutions to the complex challenges of sustainable urbanisation (Kitchin, 2014; Townsend, 2013). Conceptually, the smart city paradigm is underwritten by the promise of sustainability, economic competitiveness, and social inclusion (Arroub et al., 2016). In pursuit of these ideals, the modern city has become a living laboratory, decoding the urban environment into actionable intelligence through local testbeds and technology incubators. Yet this techno-utopian approach to urban management relies on a process of transduction — the translation of the physical city and its populace into programmable code. In doing so, technology transcends its role as a mere administrative tool: the algorithms embedded in these experiments function as governing mechanisms in their own right, allowing states and corporations to exercise power and, in Miller and Rose's (1990, p. 18) formulation, "govern at a distance" through the very infrastructure of urban life.

Despite the pervasive integration of artificial intelligence and urban analytics into public policy — predictive-policing deployments, algorithmic welfare-eligibility scoring, facial-recognition surveillance of public space — democratic oversight has lagged behind. As Pasquale (2015, p.190) warns, society is skewed to the advantage of "black box insiders," leaving the populace ignorant of how its key institutions actually function. Driven by a desire for fiscal prudence, governments are increasingly tempted to replace expensive human decision-making with cheap, supposedly objective computational logic (the Netherlands' SyRI welfare-fraud system, ruled unlawful in 2020 (Rechtbank Den Haag, 2020); Australia's 'Robodebt' scheme, condemned by a Royal Commission in 2023 (Royal Commission into the Robodebt Scheme, 2023)). Treating algorithmic management as politically neutral, however, risks sliding into "dataism" (Dijck, 2014, p. 198) — in which the socio-political dynamics of governance are sidelined entirely. This asymmetry — between the state's rapidly expanding algorithmic capacity to act upon citizens and the capacity of democratic institutions to understand, audit, and contest those actions — is what this dissertation terms the *oversight gap*.

1.2 The Dual-Use Problem

This oversight gap becomes a pressing policy crisis because the algorithmisation of civic management is inherently a dual-use enterprise. The term is used here in a stronger sense than the familiar ambiguity of dual-use tools: dual-use names a structural property of information goods, for which transfer between welfare and warfare requires not new procurement but only a change of inputs (§2.2). The dynamic is vividly illustrated by the UK government's award of the £330 million NHS Federated Data Platform contract to Palantir Technologies — a defence contractor originally backed by the CIA's In-Q-Tel, whose foundational software was forged tracking insurgent networks in Iraq and Afghanistan, and which now provides the ontological architecture for civilian health management. The case demonstrates the frictionless mobility of software as an 'information good' (Arrow, 1962, pp. 609–610): because algorithmic models possess a near-zero marginal cost of reproduction, the data ontology built to map hostile actors on a battlefield can be retrofitted to manage hospital backlogs and patient flows. When military-security apparatuses become embedded in civilian infrastructure, the governmentality of the state is structurally altered: the distinction between citizen and target, patient and pattern-of-life, collapses into a single data ontology administered by a security contractor whose institutional incentives and culture were forged in counterinsurgency operations.

The same mechanism operates in reverse: civilian urban technology is seamlessly weaponised in conflict zones to execute systemic spatial destruction — seamlessly in the economic sense, since retargeting an analytics system built for urban management demands reparameterisation rather than new procurement or physical infrastructure. For instance, Michael (2007) showed how a strategically helpless civilian political echelon abdicated responsibility for charting a political endgame to a military claiming "epistemic authority" — the power to define what counts as true knowledge about a conflict (§2.5). Today that authority is intensely algorithmic, exercised through data analytics, machine-learning target generation, and geographic information systems rather than staff papers.

The advent of Big Data reinforced the positivist assumption that massive datasets allow correlation to supersede causation — that the data can "speak for [itself]," rendering theory obsolete (Anderson, 2008, p. 1). In the ongoing systemic destruction of Gaza, the military utilises the "God's Eye View" (Sadowski, 2026, p. 269) of remote sensing, machine learning, and GIS — the very tools of peacetime urban planning — to assert epistemic dominance. Civilian politicians are drawn to the apparent objectivity of AI-driven target generation and automated damage assessment. The conceptual loop is self-reinforcing: civilian urban technology feeds military targeting through frictionless, near-zero-cost transfer, while counter-initiatives attempt to break the loop by re-injecting legal and theoretical interpretation into the data. Beneath this loop sits a deeper asymmetry, generalisable beyond any single conflict: algorithms are optimising mechanisms built to solve tactical problems — locating a target, engineering an evacuation grid, clearing a designated percentage of territory — and they cannot formulate a political resolution. The Gaza case illustrates with stark clarity a dynamic that is structural, not contingent. Nor is the pattern confined to a single conflict: in Kherson, Ukrainian civilians at markets and bus stops have been hunted by short-range drones — attacks the UN Independent International Commission of Inquiry on Ukraine has characterised as crimes against humanity (UN Independent International Commission of Inquiry on

Ukraine, 2025). The same consumer-grade drones and mapping software that optimise urban logistics define that battlespace.

As Forensic Architecture's counter-forensics demonstrate, spatial data cannot in fact speak for itself. To translate the correlation of airstrikes and destroyed medical infrastructure into the truth of domicide, one must bring the theoretical and legal framework of International Humanitarian Law (IHL) to bear on what the data indicates. In the case of the Israel Defense Forces' (IDF) conduct since October 2023, the data indicates the systematic destruction of the conditions of urban life (Forensic Architecture, 2023).

Addressing this threat in public policy requires opening the black box of dual-use data analytics. It requires understanding how the literature has tracked the shift from traditional institutional, disciplinary governance to algorithmic societies of control (Deleuze, 1990), and how the tools of urban optimisation have become simultaneously the instruments of urbicide. Deleuze's observation, made in 1990, reads today as a description of the datafied city:

> "The numerical language of control is made of codes that mark access to information, or reject it. We no longer find ourselves dealing with the mass/individual pair. Individuals have become 'dividuals,' and masses, samples, data, markets, or 'banks."' (Deleuze, 1990, p. 5)

1.3 Research Questions and Structure

Building on this Deleuzian diagnosis, this dissertation contends that the transition from disciplinary to algorithmic urban governance must be investigated systematically. Rather than arguing the dual-use claim from selected cases, it maps how three and a half decades of academic and policy literature have — and have not — registered the connections between algorithmic urban governance and spatial destruction. Two research questions structure the inquiry:

RQ1: How has the academic and policy literature from 1990 to 2025 conceptualised the dual-use nature of algorithmic technologies in urban governance, specifically in contrast to traditional physical infrastructure?

RQ2: To what extent does the evolving discourse reflect a shift from institutional (disciplinary) management of urban space to algorithmic (modulatory) control, and how is this shift implicated in instances of systemic urban destruction or domicide?

To answer them, the dissertation conducts a systematic literature review (SLR) of 275 records, following a modified PRISMA protocol (Page et al., 2021) across two complementary search strategies: one mapping the civil–military urban technology nexus, the other isolating literature on spatial destruction and the datafied city. The review is guided by a theoretical framework assembled from five sources — Arrow's economics of information goods (1962), Deleuze's societies of control (1990), DeLanda's machinic assemblages (1992; 2014), Michael's epistemic authority (2007), and the spatial-destruction theses of King, Graham, and Weizman (2004; 2006; 2007). Its most consequential finding exceeds that framework: the category of 'dual-use' itself operates inside the destruction literature as a juridical warrant for striking civilian infrastructure (§5.3).

The dissertation proceeds as follows. Chapter 2 assembles that theoretical framework and states the structural dual-use claim the review tests. Chapter 3 details the SLR methodology: search design, screening, quality appraisal, and thematic extraction. Chapter 4 presents the findings as a systematic map of the literature. Chapter 5 discusses the findings against the framework, including the review's central observation that the two searched literatures barely intersect. Chapter 6 concludes with limitations and a research agenda.

The stakes for policymakers are concrete. When a municipal government procures a GIS platform or an AI-driven traffic management system, it is — knowingly or not — acquiring technology governed by the economics of the information good; the same system, transferred to a military context, becomes a targeting apparatus through reparameterisation rather than new procurement. The policy question is not whether such transfer can happen, but whether the frameworks of oversight, procurement, and accountability that govern urban technology have registered the possibility at all. This dissertation's contribution is diagnostic before it is prescriptive: an evidence-based map of where the policy conversation has looked, and where it has been blind.

The contributions are twofold. Theoretically, the dissertation extends Arrow's economics of information goods into a named mechanism — frictionless cross-domain transfer — that the dual-use literature has documented for decades without theorising (§2.2). Empirically, it delivers a systematic map of the junction between algorithmic urban governance and spatial destruction — and shows that junction to be nearly empty (Chapter 4).

Chapter 2

Theoretical Framework

2.1 Assembling the Framework

This chapter does not survey the empirical literature on smart cities, surveillance technologies, or urban warfare — that task falls to Chapter 4. Nor does it describe the methodology by which that literature was collected — that is the function of Chapter 3. What this chapter does is assemble a deliberately selected theoretical toolkit: five concepts, drawn from economics, philosophy, military studies, and critical urban theory, that taken together enable the dual-use claim the systematic literature review will test.

That claim is specific: algorithmic governance tools are structurally dual-use in a way that physical infrastructure is not. This is not a truism about all technology. It rests on five distinct supports — economic, institutional, spatial, epistemic, and empirical — each developed in its own section below. Each contributes a necessary element; without any one the framework collapses into either technological determinism or an undifferentiated anti-surveillance critique. The framework is not presented as a definitive synthesis — it is an assembled lens whose usefulness the review is designed to test.

For the purposes of this dissertation, *dual-use* is defined as follows: a technology developed for civilian urban governance that is structurally transferable to military urban destruction with minimal modification, where the transfer is facilitated by the economic properties of software, the institutional logic of control societies, and the epistemic authority of algorithmic systems. This definition excludes generic technology dual-use (GPS, drones, the internet itself), focusing on the specific convergence of urban data infrastructure and military spatial operations.

2.2 Arrow: Information Goods and Structural Dual-Use

Kenneth Arrow's (1962) analysis of information as an economic good identifies three properties that distinguish it fundamentally from physical commodities. First, information is non-rivalrous: one actor's use does not deplete another's. A geographic information system (GIS) dataset compiled for municipal traffic management remains fully available for military targeting without degradation. Second, information is characterised by high fixed costs of production but near-zero marginal costs of reproduction: once an algorithm has been developed, its copying and redeployment is essentially costless. Third — and most consequentially — algorithmic information goods exhibit a further structural property that Arrow did not himself enumerate: frictionless transfer across domains. Because non-rivalry and near-zero reproduction costs mean the same system can address multiple problems without new production — only new parameterisation — an algorithm developed for one purpose requires minimal adaptation for another. This extension of Arrow is the framework's own theoretical contribution, and the review tests indirectly whether the literature has registered it.

These properties constitute the structural basis of the dual-use claim. Contrast them with physical infrastructure: a truck fleet built for civilian logistics can serve military purposes, but only through physical retrofitting, retraining of personnel, and redeployment of material assets — substantial and visible economic friction. An algorithm requires none of these. The same routing model that optimises waste collection in a smart city pilot can be reparameterised for targeting supply lines in a conflict zone with no new procurement — only a change of inputs. Integration, validation, and accreditation costs are not zero, but they are costs of redeployment rather than reproduction, and modest beside the friction of converting physical assets.

The dual-use literature in urban studies tends to treat dual-use as a contingent feature of specific technologies (facial recognition, predictive policing) rather than as a structural property of software itself; the SLR tests whether the literature has registered this distinction.

The Arrow extension also has a cognate in innovation economics: the general-purpose-technology literature (Bresnahan and Trajtenberg, 1995) shows that cheap reproduction still demands 'co-invention' in each new domain of application — complementary investment in integration, data, and practice — which names precisely the residual cost separating cheap reproduction from frictionless application.

2.3 Deleuze: From Discipline to Control

If Arrow explains that software dual-use is structurally low-cost and frictionless, Deleuze explains why the institutional environment is uniquely receptive to it. His 1990 *Postscript on the Societies of Control* describes a rupture in the logic of power that has direct implications for how algorithmic governance operates.

Foucault (1975) located disciplinary societies in the eighteenth and nineteenth centuries, reaching their height at the outset of the twentieth. Disciplinary power operated through vast spaces of enclosure — the prison, the barracks, the factory, the school — each with its own laws, its own spatial boundaries, its own visible architecture of control. The individual passed from one closed environment to another, and the analogy between them was explicit: "at the sight of some laborers, the heroine of Rossellini's *Europa '51* could exclaim, 'I thought I was seeing convicts"' (Deleuze, 1990, p. 3).

Deleuze argues that these enclosures are in a state of generalised crisis. We have moved from societies of discipline to societies of control, and the distinction is not merely technological but logical. Enclosures are moulds, distinct castings; controls are a modulation, "like a self-deforming cast that will continuously change from one moment to the other" (Deleuze, 1990, p. 4). Disciplinary power was punctual — it began and ended at the gates of the institution. Control is continuous, ambient, and always-on. The factory disciplined its workers during shifts; the corporation modulates its employees through perpetual assessment, continuous feedback loops, and algorithmic performance metrics that follow them home.

First, the *dividual*: individuals become "dividuals" (Deleuze, 1990, p. 5), decomposed into data points — locational, behavioural, biometric — each of which can be tracked, scored, and acted upon independently. This is the ontological precondition for DeLanda's targetable city: you cannot target an individual, but you can target a dividual's data signature.

Second, the *code*: control societies operate not through watchwords (the sign of disciplinary integration and resistance) but through passwords — "codes that mark access to information, or reject it" (Deleuze, 1990, p. 5). Urban governance in a control society functions through algorithmic access and denial: the smart card that grants or refuses entry, the credit score that forecloses housing, the predictive policing algorithm that saturates one neighbourhood with patrols while leaving another unmonitored. These are not visible walls but continuous modulations of access and restriction.

Third, Deleuze cites Guattari's vision of a city where one would leave one's apartment, one's street, one's neighbourhood thanks to an electronic card — "but the card could just as easily be rejected on a given day or between certain hours; what counts is not the barrier but the computer that tracks each person's position — licit or illicit — and effects a universal modulation" (Deleuze, 1990, p. 7). This is the smart city *avant la lettre*, and it illustrates why algorithmic dual-use is harder to regulate than physical dual-use: regulation conventionally targets visible institutional boundaries (the wall, the checkpoint, the procurement contract), not continuous flows of data and modulation.

Modulation is what makes algorithmic dual-use architectural rather than accidental. The same urban sensor network can optimise traffic flow and enforce movement restrictions, because both are parameter changes in the same system. No institutional rupture is required — only a modulation.

2.4 DeLanda: Machinic Assemblages and the Targetable City

Deleuze supplies the logic of control; Manuel DeLanda supplies its terrain: the city itself, reconceived as a programmable field.

In *War in the Age of Intelligent Machines* (DeLanda, 1992), DeLanda traces the progressive automation of military decision-making and the emergence of the battlespace as a machinic assemblage — a heterogeneous network of human and non-human components (sensors, algorithms, command structures, weapons platforms) that functions as a directed system. The battlespace is not a given terrain but a constructed one: it must be mapped, datafied, and rendered machine-readable before it can be fought over — and the same is true of the city.

DeLanda's broader philosophical project, developed in *Intensive Science and Virtual Philosophy* (2014), supplies the ontology: drawing on dynamical systems theory, he replaces essentialist categories with multiplicities — possibility spaces defined by attractors and bifurcations. Applied to the city, this replaces a fixed entity with essential properties (residential, commercial, industrial) with a state space: a field of possible configurations structured by the data infrastructure that models it.

Read through DeLanda, the city becomes a targetable field — a space where the same population-level data (density, flow, infrastructure networks, resource distribution) functions simultaneously as an optimisation input for civilian governance and a targeting matrix for military destruction. The smart city's sensor network maps traffic congestion for optimisation; the same network maps population density for targeting. The GIS platform that zones a neighbourhood for mixed-use development shares its core software architecture with the GIS platform that designates it as an evacuation grid. The city becomes, in effect, a single substrate, parameterised for welfare or warfare.

Arrow explains why the transfer is cheap; Deleuze explains why institutions are receptive to it; DeLanda explains where it happens — in the datafied urban fabric itself.

2.5 Michael: Epistemic Authority Through Data

DeLanda shows us how the city becomes a programmable field. Kobi Michael shows us who gets to program it — and why that authority is structurally dangerous. Michael (2007) concept of the IDF as an epistemic authority is the most directly applicable account of how algorithmic governance translates into political power. Michael argues that during the era of Low Intensity Conflict, the Israeli political echelon — experiencing "strategic helplessness" (p. 428) and operating with "creative fuzziness" (p. 429) — effectively abdicated its responsibility to chart a political endgame. Into this vacuum stepped the IDF, claiming not merely operational capability but epistemic authority: the right to define what counts as true knowledge about the conflict.

The mechanism was systematic knowledge production. Under Ya'alon's leadership, the IDF developed an elaborate Estimation of the Situation (EOS) process — knowledge maps, conceptual frameworks, staff methodologies — that far exceeded the civilian government's analytical capacity. Cabinet discussions, lacking alternative knowledge infrastructures, invariably reverted to the tactical and operational level, "lacking sufficient depth" (Michael, 2007, p. 439). The military dominated the "discourse space" (p. 444) through informational dependence rather than coercion: the political echelon came to rely on military knowledge as the only available framework for understanding the conflict — in Foucault's terms, the IDF became the "truth and lie agent" (p. 444), the institution that determines which propositions enter the discourse and which are excluded.

This dissertation updates Michael's framework in three ways. First, the algorithmic turn: epistemic authority is now exercised not primarily through linguistic dominance and staff work but through data analytics, AI-driven target generation, and GIS — and is harder to contest because it appears objective and computational. When a targeting recommendation arrives as the output of a machine-learning model rather than as a general's argument, it carries the veneer of computational neutrality that Pasquale (2015) names the black-box problem (§1.1).

Second, the material upgrade: Sadowski (2026, p. 269) discussion of the "God's Eye View" (Haraway, 1988) describes how remote sensing and satellite surveillance give the military an epistemic monopoly on spatial truth: only the military can produce the 'ground truth' against which all other claims are measured. This monopoly is materially underwritten by the same satellite constellations and sensor arrays that underpin civilian urban management.

Third, collateral penetration: the same sensor networks that produce civilian urban data — traffic flows, zoning maps, population density estimates — feed seamlessly into military intelligence. The dual-use convergence is epistemic before it is operational: before a single targeting decision is made, the epistemic framework has already been set by data infrastructures that are structurally indifferent to their application. Michael's informational dependence is now built into the data infrastructure itself.

2.6 King, Graham, and Weizman: Domicide, Urbicide, Vertical Sovereignty

The framework so far is a chain of mechanisms — economic, institutional, spatial, epistemic. Three thinkers show where that chain ends: King names what is violated, Graham describes the geopolitical architecture of its violation, and Weizman documents the limit-case.

King (2004) provides the phenomenological ground. In *Private Dwelling*, King argues that the home is not merely a physical structure but an ontologically defined interior space — the site of privacy, boundary, and dwelling: where the individual is most fully constituted as a private subject, shielded from the gaze of the state and the market. Algorithmic penetration of the home — through GIS mapping, predictive policing, smart city sensor networks — is not merely surveillance but an ontological violation of dwelling. The dual-use implication is direct: the same data infrastructure that optimises housing allocation, identifies areas of housing stress, and models population movement can identify homes for demolition, populations for displacement, and neighbourhoods for erasure.

Graham (2006; 2011) provides the geopolitical architecture. His concept of the "new military urbanism" (2011, p. 1) describes the systematic reframing of cities as the primary battlespace of the twenty-first century. Urban infrastructure — water systems, electricity grids, telecommunications networks — is no longer collateral damage but the target itself, deliberately struck to achieve political effects through the demodernisation of urban life. Graham's concept of vertical sovereignty describes how drones, satellite surveillance, and aerial bombing change the geometry of urban conflict: control is exercised through volumetric dominance of the airspace rather than territorial occupation.

Crucially, Graham documents how civilian urban management systems — the very technologies of the smart city — are co-opted for military operations: for Graham the dual-use claim is not theoretical but empirically documented. His discussion of "dual-use electrical infrastructure" — where the US military's deliberate degradation of Iraq's power grid during the 1991 Gulf War produced "a form of urbicide — the systematic destruction of the modern city" (Graham, 2011, pp. 1, 355) — shows the same logic that resurfaces later in the review as a juridical category (see §5.3), where ‘dual-use’ shifts from an analytic descriptor to an operational targeting warrant.

Weizman (2007) provides the limit-case. *Hollow Land* analyses the Israeli occupation of Palestinian territory as a laboratory for vertical sovereignty, datafied spatial control, and the weaponisation of urban planning. Weizman shows how architecture, zoning law, and infrastructure planning function as instruments of domination: the settlement wall, the checkpoint, the bypass road each function as spatial technologies of control whose logic the algorithmic systems discussed above inherit and intensify.

The work of Forensic Architecture (2023–present), which Weizman founded, extends this analysis into the era of AI-driven targeting: their documentation of the systematic destruction of Gaza since October 2023 — clearing territory, engineering corridors, implementing “evacuation grids” — illustrates the systematic weaponisation of urban data infrastructure. The “evacuation grid” is a municipal zoning tool (a GIS overlay dividing urban space into numbered sectors) weaponised for mass displacement: a civilian planning instrument repurposed as an instrument of domicide with minimal modification. Together they show that the framework is not abstract: it describes outcomes that are already observable.

2.7 Synthesis: The Analytical Lens

The framework generates a structural claim that the SLR is designed to test:

> The dual-use nature of algorithmic governance is not incidental or sectoral. It is structurally determined by the economic properties of information goods (Arrow), institutionalised by the transition from disciplinary to control logic (Deleuze), embedded in the urban fabric as a programmable field (DeLanda), legitimised through data-driven epistemic authority (Michael), and terminates in spatial destruction — domicide, urbicide, vertical sovereignty — where that authority operates unchecked (King/Graham/Weizman).

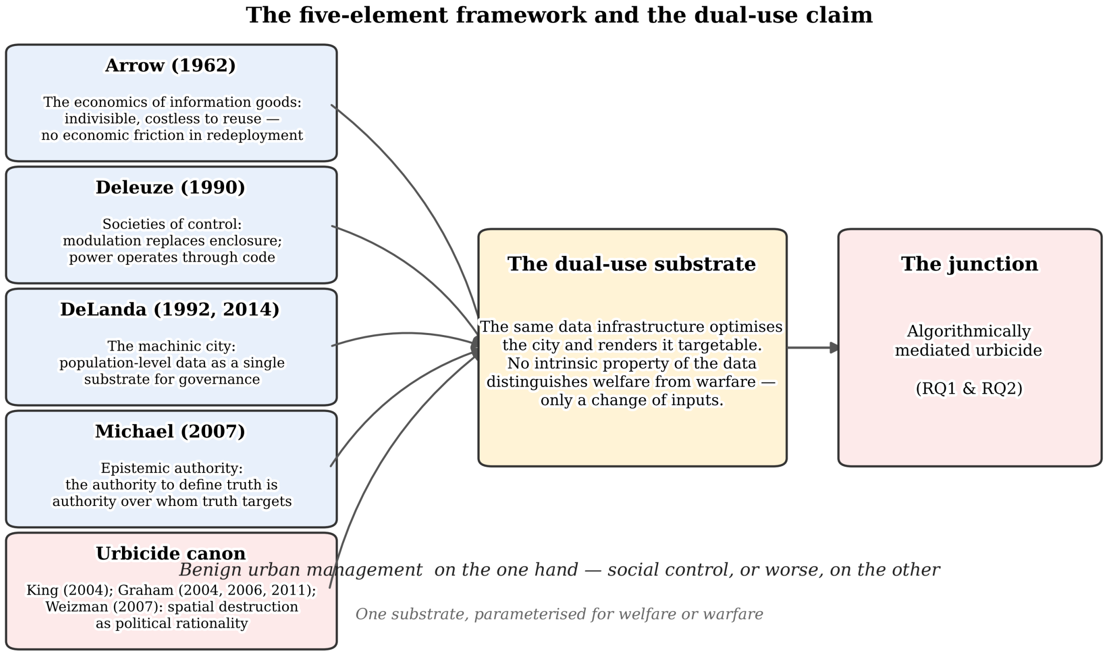

Figure 2.1: The five-element framework and the dual-use claim. Arrow's economics of information goods, Deleuze's societies of control, DeLanda's machinic city, Michael's epistemic authority, and the urbicide canon (King, Graham, Weizman) converge on a single data substrate — one that can be parameterised for welfare or warfare with no change to the infrastructure itself.

This is a strong claim, and it is important to be precise about what the SLR can and cannot do with it. The review does not test the claim's truth — that would require a different methodology. What it tests is whether the literature has registered the connections the framework proposes. Do researchers working on smart cities and algorithmic governance engage with the spatial destruction that these tools enable? Do scholars documenting urbicide and domicide engage with the data infrastructures that facilitate it? Does the literature connect the economic mechanism that enables dual-use to the spatial outcomes it produces — or are these treated as independent domains?

The preliminary evidence from the review's search phase suggests the latter. The two search strategies deployed in this SLR — Search A (civil-military urban technology nexus, high recall) and Search B (spatial destruction and the datafied city, high precision) — returned zero overlapping records at the screening stage: of 874 unique records screened and 151 included, not one appeared in both strategies. The extraction results reinforce this separation across the coded corpus (Chapter 4): the governance side scores high on the framework's first three themes but is virtually silent on spatial destruction, while the destruction side shows the inverse pattern. The junction the framework predicts — algorithmically mediated spatial destruction — is documented in only five records in the entire coded corpus of 270 (§4.5.4).

This separation is itself a finding. If the framework assembled in this chapter is coherent — if the arc from Arrow through Deleuze, DeLanda, and Michael to King, Graham, and Weizman describes a real structural dynamic — then the literature's failure to trace that arc is not merely an academic gap. It is an epistemic problem with direct policy consequences: the people who design urban data systems and the people who document urban destruction are not reading each other. The SLR's contribution is to make that failure visible, systematic, and actionable.

Chapter 3

Methodology: A Systematic Mapping of the Literature on Algorithmisation, 1990–2025

3.1 Review Design

To empirically investigate the intersection of algorithmic governance, urban space, and systemic spatial destruction (domicide), this dissertation employed a Systematic Literature Review (SLR). Where a traditional narrative review risks selection bias by design — the 'cherry-picking' of sources to validate a pre-existing critique — the SLR provides a transparent, auditable, and reproducible mechanism for data collection. This aligns with the 'evidence turn' in contemporary public policy and management, which increasingly demands structured methodologies to track policy evolution and evaluate 'what works' (Tranfield et al., 2003). Rather than evaluating the effectiveness of a specific policy intervention, however, the review was adapted to rigorously map a longitudinal discursive shift: how global academic discourse and policy literature conceptualised algorithms, big data, and spatial surveillance over three and a half decades.

The legitimacy of this design rests on an established methodological lineage. Developed in the health sciences, the SLR's protocol-driven design converts the literature review from an essayistic exercise into an auditable instrument: explicit search strings, pre-registered inclusion criteria, and logged screening decisions make the reviewer's judgements visible and reproducible (Sutton et al., 2019). Within Sutton and colleagues' taxonomy of the 'review family', this is a *mapping* review: it trades depth for breadth, charting the structure and distribution of a heterogeneous literature — exactly the analytical object this dissertation requires, since its claims concern the *configuration* of a field rather than the pooled effect of an intervention (Sutton et al., 2019). The approach carries a direct precedent in social policy: Gurney's (2023, pp. 239–241) systematic literature mapping of 213 COVID-19 housing studies demonstrates the method's value for converting an unwieldy, fast-moving policy literature into structured evidence of what the discourse has and has not registered. The present review applies the same logic to a colder question: not what the literature says, but where it has been silent.

The review was conducted between June and July 2026; every stage was logged with timestamps, record counts, and tooling versions, and is reported against the PRISMA 2020 statement (Page et al., 2021) (Figure 3.3), rendering the final corpus fully auditable. The complete review archive — search records, screening verdicts, coding outputs, pipeline scripts, and the timestamped SLR log — is publicly available at `https://github.com/EVasiloudes/MSc-PPM-Dissertation---Repo`.

3.2 Conceptual Scope and Research Questions

The core premise motivating the search parameters is the hypothesis that the algorithmisation of public policy possesses a dual-use nature that distinguishes it from the management of physical urban infrastructure. Where the build-out of legacy urban assets — highways, rail networks, gas terminals — requires massive capital expenditure, physical friction, and explicit geographic constraint, software operates under entirely different economic mandates: following Arrow's (1962, pp. 609–610) analysis of the 'information good' (developed fully in Chapter 2), a high initial cost of development but a near-zero marginal cost of reproduction and redeployment. The review was designed to test whether — and how — this phenomenon has been recognised, debated, or obfuscated in the literature.

The research questions are stated in full in Chapter 1 (§1.3). Operationalised for the review, they required the collected literature to speak to two things at once: the dual-use nature of algorithmic technologies in urban governance, specifically in contrast to physical infrastructure (RQ1); and the shift from institutional, disciplinary management of urban space to algorithmic, modulatory control, insofar as that shift is implicated in systemic urban destruction or domicide (RQ2). These questions fixed the perimeter of the search.

3.3 Search Strategy

3.3.1 Databases and date range

The review interrogated a 35-year chronological window, 1990–2025. This expansive range was a methodological necessity, allowing the review to track the literature across distinct technological epochs: from the early internet and the digitisation of municipal records in the 1990s, through the Big Data era of ubiquitous harvesting, to the current period defined by generative AI and autonomous systems. The window's breadth sacrifices depth at its edges — the early-1990s digitisation literature is sparse, and pre-1990 debates on automated warfare fall outside it entirely — but buys symmetric coverage of both literatures' full trajectories; the corpus's own distribution confirms the wager, with the post-2015 concentration of included records (Figure 3.1) showing that the window captures the period in which the two vocabularies begin to converge.

Peer-reviewed literature was sourced from Web of Science and Scopus, chosen for their breadth across the social sciences, urban planning, and defence studies. Grey literature was sourced from Policy Commons, supplemented by targeted manual searches of key institutional repositories (RAND Corporation, Chatham House, Urban Displacement Project), emphasising policy documents, parliamentary records, defence white papers, and think-tank outputs addressing the dual-use implications of urban data technologies.

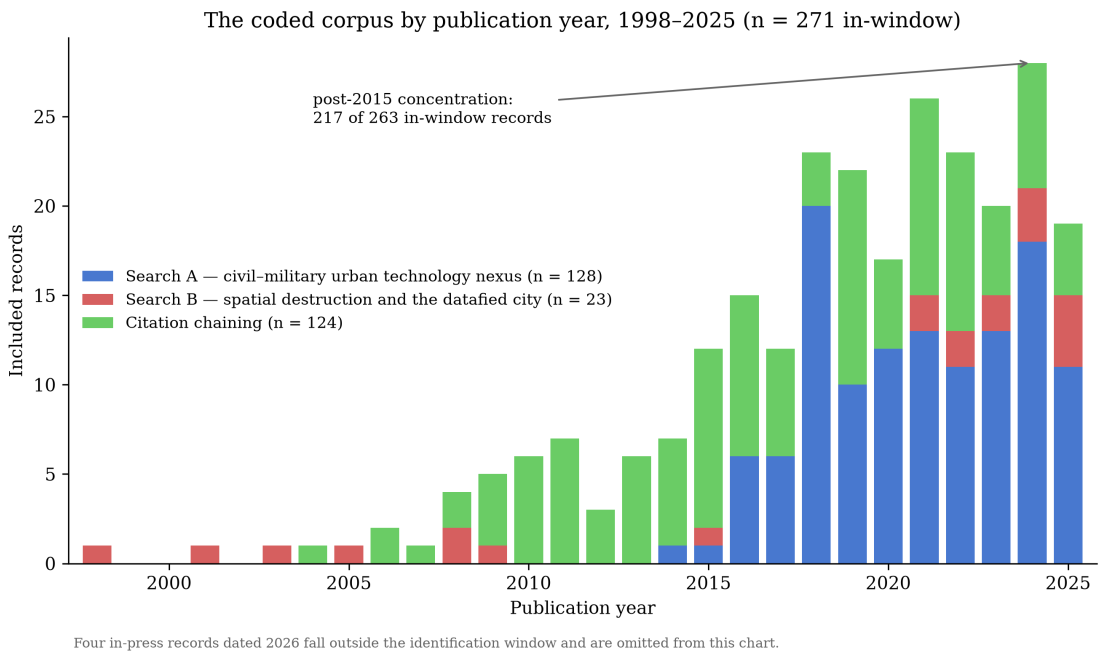

Figure 3.1: Publication-year distribution of the included corpus (n = 271 in-window records), stacked by provenance. The concentration of records after 2015 — driven by the smart-city wave on the governance side and by post-2021 work on the destruction side — reflects the technological epochs the window was chosen to span; four in-press records dated 2026 fall outside the identification window.

The window's bookends are corroborated by field-level publication trends (Figure 3.2): the governance vocabulary scarcely exists in the indexed literature before the mid-2000s, while the destruction vocabulary long predates the window and persists at a low, steady volume throughout it. A 1990 start therefore captures the full arc of the algorithmic-governance literature from its prehistory, and the 2025 end-point coincides with the moment the destruction literature begins to register algorithmic systems directly.

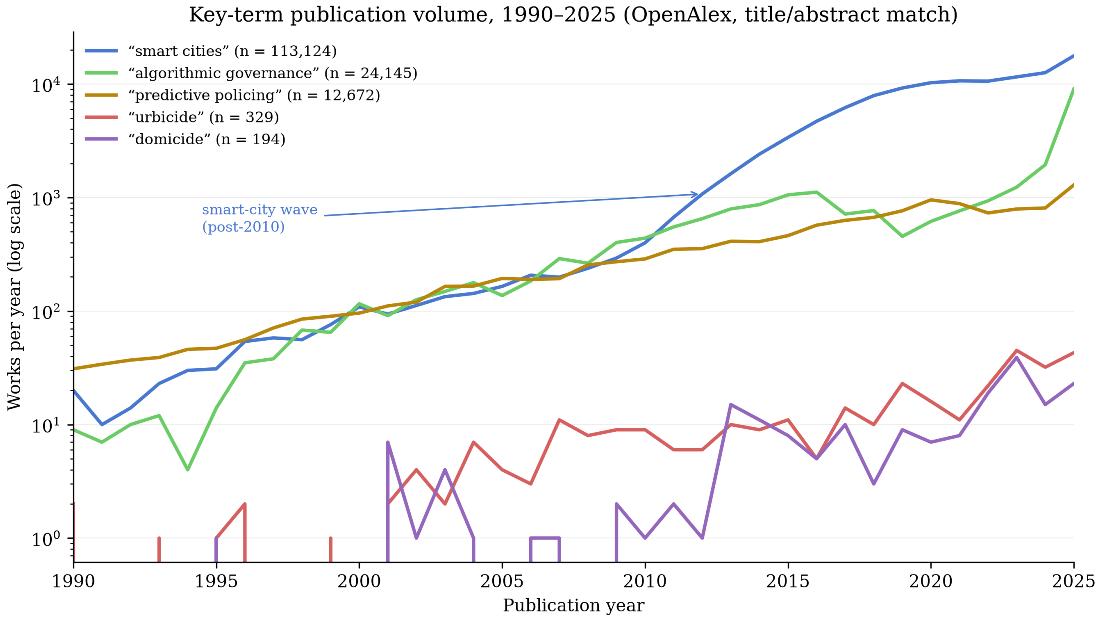

Figure 3.2: Publication volume for five key terms, 1990–2025, log scale (OpenAlex works matching each term in title or abstract; retrieved 6 August 2026). The governance terms ("smart cities", "algorithmic governance", "predictive policing") exhibit the post-2010 spike the window was designed to capture; the destruction terms ("urbicide", "domicide") remain two orders of magnitude smaller, foreshadowing the asymmetry documented in Chapter 4.

3.3.2 Dual-search design

Given the cross-cutting nature of the inquiry — bridging urban analytics, security studies, and spatial destruction — a single search string risked being either too restrictive or analytically shallow. The review therefore deployed two complementary search strategies, merged at the deduplication stage.

Search A — Civil–military urban technology nexus (high recall): designed to surface the broader landscape of how algorithmic governance, smart city infrastructure, and urban analytics intersect with military, surveillance, and security apparatuses; the string pairs an algorithmic-governance vocabulary ("algorithmic governance", "urban analytics", "smart cit*", "platform urbanism", "predictive policing", "urban big data", "data-driven urbanism") with a security vocabulary ("dual-use", "civil-military", militari*, "national security", securitis*/securitiz*, weaponi*, warfare, "state/mass surveillance", "OSINT").

Search B — Spatial destruction and the datafied city (high precision): designed to capture literature specifically addressing how algorithmic and data-driven tools are implicated in urbicide, domicide, and spatial violence; the string pairs destruction terms ("domicide", "urbicide", "spatial violence", "home unmaking", "urban destruction", "urban warfare") with data-technology terms ("data", "algorithm", "AI", "surveillance", "GIS", "remote sensing"). Platform-specific formulations are reproduced verbatim in Appendix B.

This dual-search approach ensured that papers theorising the surveillance–governmentality nexus (which might not explicitly invoke 'domicide') and papers documenting spatial destruction (which might not deploy the language of 'algorithmic governance') were both captured. The convergence — or absence of convergence — between the two literatures was itself a phenomenon the review was designed to surface.

3.3.3 Calibration

Search strings were calibrated through documented scoping iterations. An initial Scopus formulation of Search A returned 9,981 records — unmanageable for single-author screening. Diagnosis identified four generic terms driving cross-disciplinary noise — 'big data', bare 'security', bare 'targeting', and bare 'surveillance' — which were replaced with the qualified equivalents and truncated stems shown above; a date-boundary error that would have excluded 1990 and 2025 was also corrected.

Policy Commons required separate calibration. Initial strings returned 13,096 (Search A) and 3,031 (Search B) hits, owing to automatic stemming and full-text indexing across a broad grey corpus; both searches were revised to fielded formulations (Search A: both legs in the summary field; Search B: spatial-destruction terms in title, data terms in summary), returning 63 and 10 exportable records respectively.

All searches were executed on 3 July 2026.

3.3.4 Citation chaining

The second instrument works not through keywords at all but through the literature's own citation web — a procedure known as citation chaining, or 'snowballing' (Wohlin, 2014). The logic is that relevant work the keyword searches missed — because it speaks a different disciplinary vocabulary — will still cite, or be cited by, the field's canonical texts. One iteration of chaining was therefore executed from four anchor texts: Graham (2006), Kitchin (2014), and Weizman (2007), plus Michael (2007), which is central to the theoretical framework (*forward* chaining harvests later works citing an anchor; *backward* chaining harvests the anchor's own reference list). The pass was executed via OpenAlex (Priem et al., 2022) between 19 and 28 July 2026, using the same date range (1990–2025), language (English), and document-type filters as the database searches.

Backward chaining was possible only for the two journal-article anchors (Kitchin, 2014: 33 filtered references; Michael, 2007: 8 filtered references), because OpenAlex indexes no reference lists for the two monographs (Graham, 2006; Weizman, 2007). Forward chaining returned substantially more material: Kitchin (1,492 citing works), Weizman (346), Graham (192), and Michael (17). Deduplication against the main corpus removed 9 duplicates from a raw pool of 2,090 candidates, leaving 2,081 unique records (7 records arrived via multiple anchors and are counted once).

Snowball candidates were enriched for missing abstracts via the Springer META API (217 of 225 Springer DOIs recovered) and an OpenAlex re-scan (18 of 224 remaining DOIs recovered). Records with no retrievable abstract (242) were excluded at triage, mirroring the main pipeline rule. The remaining 1,839 records were screened by title and abstract using the same AI-assisted workflow, prompt, model, and parameters as the database searches (DeepSeek V4 Flash; temperature 0.0; §3.4.3), yielding 124 Includes, 279 Maybes, and 1,436 Excludes (1,678 including the triage exclusions).

The 279 Maybe records were dropped without manual review. They represent 13% of the snowball pool, and their titles and sources indicate predominantly tangential material: governance papers mentioning surveillance in passing, smart-city critiques without dual-use relevance, and methodological pieces about big data without policy or spatial dimensions. The 124 Includes already represent a substantial snowball corpus (82% the size of the 151-record database corpus), and the time cost of manually resolving 279 borderline records was disproportionate to the expected marginal gain. This decision is disclosed as a limitation in §3.6.

The final snowball corpus of 124 records is composed of: Weizman 2007 forward

(46), Kitchin 2014 forward (42), and Graham 2006 forward (38); neither Michael 2007 nor the backward passes yielded includes. Notably, 2 records arrived via multiple anchors. The complete pipeline, counts, and screening prompt are archived in the SLR log and `analysis/snowball/` directory.

3.4 Screening and Selection

3.4.1 Deduplication and corpus assembly

Results from all six searches were exported to Zotero for reference management and deduplication. Of 963 compiled records, 89 duplicates were removed (80 matched by DOI, 9 by fuzzy title matching), leaving 874 unique records: 698 from Search A and 176 from Search B. Notably, zero records appeared in both searches at this stage — an early signal of the discursive separation that Chapter 4 develops as a central finding.

3.4.2 Inclusion and exclusion criteria

Records were screened against written criteria fixed in advance. **Inclusion:** (a) peer-reviewed journal articles, book chapters, or authoritative theoretical texts, or (b) substantive policy documents (white papers, parliamentary reports, think-tank analyses) from identifiable institutional authors; published within 1990–2025; and explicitly addressing the intersection of data technology with public policy, civic governance, urban space, or state control. **Exclusion:** literature focused purely on the technical, mathematical, or computational mechanics of software engineering without applied reference to public policy, civic governance, the urban environment, or defence strategy; editorials, conference abstracts without full papers, and journalistic commentary; and non-English-language articles without high-quality translations (acknowledged in §3.6). **Quality appraisal:** peer-reviewed status served as the baseline threshold for academic literature; grey literature was appraised using the AACODS checklist (Authority, Accuracy, Coverage, Objectivity, Date, Significance) to filter advocacy material lacking evidentiary grounding (Tyndall, 2010).

3.4.3 Single-reviewer screening with AI assistance

As a single-author dissertation, conventional dual-reviewer screening was not feasible. Screening was therefore conducted by the author with AI assistance: a large-language-model classifier (DeepSeek V4 Flash; temperature 0.0 for deterministic output) assigned provisional include/exclude verdicts against the written criteria, working through triage priority bands (195 HIGH, 326 MEDIUM, 276 LOW, with 77 records excluded at triage for absent or vestigial abstracts). This is an experimental workflow, disclosed in the AI-Use Statement (Appendix A), and should be read as an exploration of emerging research methods rather than a settled protocol. Borderline cases were logged in a decision journal and resolved against the written criteria rather than ad hoc judgement; doctoral dissertations were excluded on manual review, while full conference papers were retained as eligible. The validity of the AI-assisted verdicts was checked through random sampling of screening results, and the same procedure was applied to the coding process described in §3.5. All final decisions remained the author's responsibility.

Two features of this workflow deserve emphasis. The first is scale: 874 database records and 1,839 snowball candidates — 2,713 screening decisions in total — represent between one and two person-months of manual screening at conventional rates; the AI-assisted protocol compressed this to hours of computation, redirecting the author's time to calibration, borderline resolution, and verification sampling. The second is precedent: machine-assisted screening is an established, peer-reviewed component of systematic-review practice (Schoot et al., 2021); what this dissertation adds is a documented account of the same division of labour with a general-purpose language model, including its failure modes.

3.4.4 Screening results

Of 874 records screened by title and abstract, 151 were included (17.2%) and 723 excluded.

Snowball screening followed the same protocol and criteria. Of 2,081 unique citation-chained candidates (242 triage-excluded for no abstract, 1,839 screened), 124 were included (6.7% of screened), 279 marked Maybe (dropped; §3.3), and 1,678 excluded. The combined corpus across both identification streams totals 275 records: 151 from database searches and

124 from citation chaining.

3.5 Full-Text Retrieval and Thematic Coding

Full texts were retrieved and managed in Zotero. Of the 151 included records, 147 yielded usable text and were coded; the four failures, together with the snowball gap below, are itemised as limitations in §3.6. Extracted texts were logged in a Systematic Review Map recording each text's publication year, discipline, geographic focus, document type, and conceptual orientation.

For the snowball corpus, 123 of 124 records required fresh PDF retrieval (only 1 overlapped with the database-search PDFs). Retrieval and coding completed on 28 July 2026: 123 of 124 snowball records were successfully retrieved, text-extracted, and coded; one record (a University of North Carolina repository chapter) yielded no retrievable PDF. The combined coded corpus therefore comprises 270 records (147 database-search + 123 snowball) from 275 included records — a coding success rate of 98.2%.

Coding combined deductive structure with inductive openness. Four themes, derived from the theoretical framework in Chapter 2, guided extraction, each scored on a 0–3 scale (absent $\rightarrow$ central):

1. **The Information Good and Dual-Use Fluidity** — recognition of the near-zero marginal cost of algorithmic tools and their frictionless translation from civilian urban technology to military application (Arrow, 1962).

2. **Epistemic Authority and Black-Boxing** — deference by civilian policymakers to the seeming objectivity of military or corporate data systems (Michael, 2007; Pasquale, 2015).

3. **The Machinic City and Modulation** — conceptualisation of urban populations and housing as dynamic, targetable data fields rather than static infrastructure (Deleuze, 1990; DeLanda, 1992; DeLanda, 2014).

4. **Spatial Collapse and Home Unmaking** — explicit connection of weaponised urban

planning and algorithmic zoning tools to displacement, urbicide, and the infringement of the private dwelling (King, 2004; Graham, 2004; Graham, 2011; Weizman, 2007).

Supplementary fields captured dual-use explicitness, direction, and structural framing, plus methodology, geography, thesis, key quotations, and policy relevance. Because the review tracks a longitudinal discursive shift, every coded theme was additionally plotted against publication date, letting the synthesis track when — and in which discourse communities — dual-use awareness emerges or disappears across the window.

Coding was executed by the author with AI assistance under the same disclosure terms as screening: the model scored papers in resumable batches against a written codebook (temperature 0.0), with per-paper JSON outputs retained for qualitative review; very long texts were sampled under a documented truncation rule (introduction, middle sections, conclusion). All coding decisions remained the author's responsibility and are disclosed in Appendix A.

3.6 Synthesis Approach and Limitations

The synthetic phase bridges the empirical literature map back to the theoretical framework of Chapter 2. Rather than a quantitative meta-analysis — inappropriate for qualitative policy discourse — the synthesis combines descriptive statistics of theme distributions with qualitative deep-dives into the small set of 'bridge' papers scoring across all four themes, testing whether the literature has registered the connections the framework proposes.

Six limitations are acknowledged. **First,** the English-language restriction risks underrepresenting discourse from non-Anglophone policy communities — particularly relevant given the geographic foci of the case material — and is flagged wherever the synthesis makes global claims. **Second,** single-reviewer screening and coding, while mitigated by deterministic AI assistance, written criteria, and decision journaling, cannot fully replicate the inter-rater reliability of team-based reviews. **Third,** grey-literature indexing is structurally uneven: classified or restricted defence documents are by definition absent, so the review maps the public policy discourse only; reliance on Policy Commons after the lapse of the Overton trial licence further constrains grey-literature coverage. **Fourth,** backward citation chaining was possible only for journal-article anchors: OpenAlex indexes no reference lists for the two monographs (Graham, 2006; Weizman, 2007), so the backward pass drew only from Kitchin (2014) and Michael (2007). **Fifth,** 279 Maybe-verdict snowball records were dropped without manual review (§3.3); some keyword-invisible relevant records will have been missed, and the 124-record snowball corpus cannot be treated as exhaustive of the citation network. **Sixth,** five of 275 included records (1.8%) could not be coded: three database-search records yielded no extractable text, one database-search record's retrieved PDF proved to be the wrong source (a 631-page edited handbook that does not contain the chapter it was catalogued as), and one snowball record had no retrievable full text. Among these is one record — a chapter on surveillance and urbicide in Xinjiang — whose subject matter is directly relevant to the review's core question; its absence biases the corpus, if anything, against the dissertation's thesis, and so is judged acceptable. **Seventh,** a post-coding audit identified seven duplicate pairs within the 151 database-search includes — record pairs that survived deduplication because one copy lacked a DOI or carried variant metadata (in one case, two versioned DOIs of the same repository item). Counted by unique publication rather than by record, the database includes number 144, the combined corpus 268, and the coded corpus 263. Because each duplicate's second copy was coded independently, theme counts are marginally affected — on a unique-publication basis, records scoring ≥ 2 number 84, 115, 99, and 74 for Themes 1–4 respectively — but the magnitude and direction of every reported pattern are unchanged, and the five-record intersection (§4.5.4) is unaffected. Counts reported throughout are the records as screened and coded. Each limitation constrains the scope of the claims the synthesis can support, and the conclusion is calibrated accordingly.

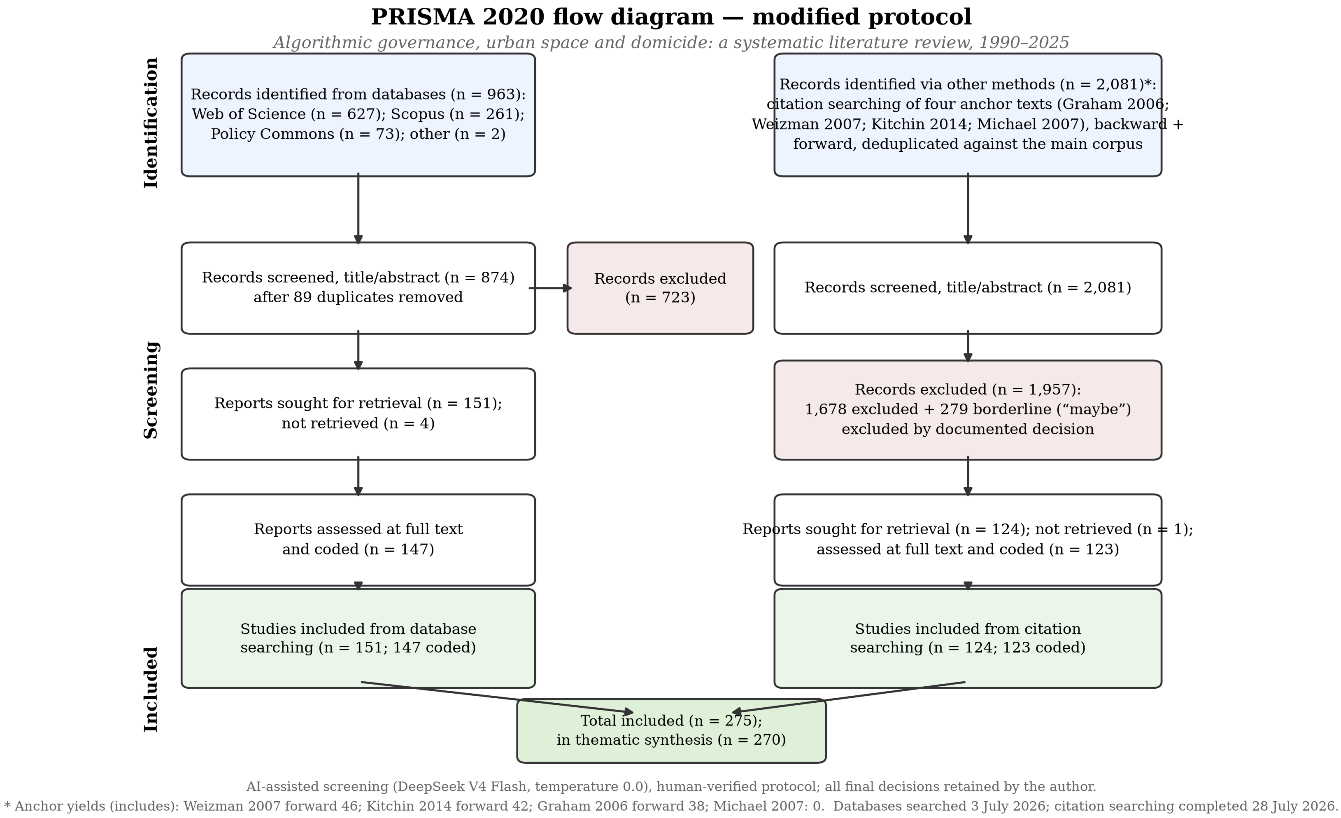

Figure 3.3: PRISMA 2020 flow diagram for the modified protocol. Final counts as of 28 July 2026: 874 unique records screened from database searches and 2,081 deduplicated citation-chaining candidates, yielding 275 included studies (270 coded).

Chapter 4

Findings: The Systematic Map

4.1 The Shape of the Corpus

The final corpus comprises 275 records: 151 identified through database searches (Web of Science, Scopus, Policy Commons) and 124 through backward and forward citation chaining on four anchor texts (Graham 2006; Weizman 2007; Kitchin 2014; Michael 2007). Full extraction and coding succeeded for 270 records (98.2%); the five failures are itemised in Section 3.6.

Three structural features of the corpus matter for what follows.

First, *recency*. The literature is largely a twenty-first-century phenomenon, and mostly a post-2010 one: 111 records date from the 2010s and 136 from the 2020s alone, against 16 from the 2000s. Publication peaks in 2024 (n=28). The convergence this dissertation examines — algorithmic governance meeting urban destruction — is being theorised now, in real time; much of the destruction-side literature concerns events still unfolding at the time of writing. The pattern holds on both sides of the corpus: 69 of 128 Search-A records and 13 of 23 Search-B records date from 2021 or later, and even the citation-chained corpus — anchored in older canonical texts — counts 38 of 124 (Figure 3.1).

Second, *genre*. Journal articles dominate (69%), but the corpus is notably hybrid: book chapters (11%), conference papers (8%), policy reports and grey literature (7%). The grey literature is not incidental. The UK Ministry of Defence's Future Cities reports, the US-China Economic and Security Review Commission's analysis of Chinese smart-city development, and NATO-adjacent technical papers on military exploitation of smart-city IoT constitute a practitioner literature that confirms, from inside the defence establishment, the feasibility claims the critical literature makes from outside (Section 4.2).

Third, *method*. The corpus skews interpretive: theoretical work (34%) and qualitative empirical work (23%) dominate, with case studies at 15% and policy analysis at 10%. Quantitative empirical work is scarce (7%). This is itself a finding: the intersection of algorithmic governance and spatial destruction has been theorised and narrated far more than measured — with the striking exception of remote-sensing damage assessments of Gaza (Holail et al., 2024), where the same sensing stack used for targeting is turned to documentation.

Table 4.1: Mean theme scores by corpus (0–3 scale; n scoring ≥2 in parentheses)

Theme	**Search A** (n=126)	**Search B** (n=21)	**Snowball** (n=123)	**Combined** (n=270)
T1: Dual-Use Fluidity	1.21 (56)	0.67 (5)	0.59 (25)	0.89 (86)
T2: Epistemic Authority	1.33 (62)	0.57 (3)	1.17 (52)	1.20 (117)
T3: Machinic City	1.33 (54)	0.86 (4)	0.99 (44)	1.14 (102)
T4: Spatial Destruction	0.32 (6)	1.81 (12)	1.33 (58)	0.90 (76)

The asymmetry in Table 4.1 is the corpus's first lesson. The keyword-search corpus and the citation-network corpus are not two samples of one literature; they are two literatures that barely cite each other. Database searching recovered the governance side — dual-use transfer, epistemic authority, the programmable city. Citation chaining from Graham and Weizman recovered the destruction side — urbicide, domicide, home unmaking — which keyword searching had largely missed (only 18 of 147 main-corpus records score ≥2 on Theme 4, against 58 of 123 snowball records). The discursive separation first visible at deduplication — zero records appearing in both searches (§3.4) — is reproduced at corpus scale. Michael (2007), notably, yielded no included records at all: the epistemic-authority literature it anchors was already captured by the Weizman and Graham networks, and the counterinsurgency-theory branch does not intersect the urban-analytics branch. The literatures of how algorithms govern and how cities are destroyed run in parallel. Mapping where they meet is the task of this chapter.

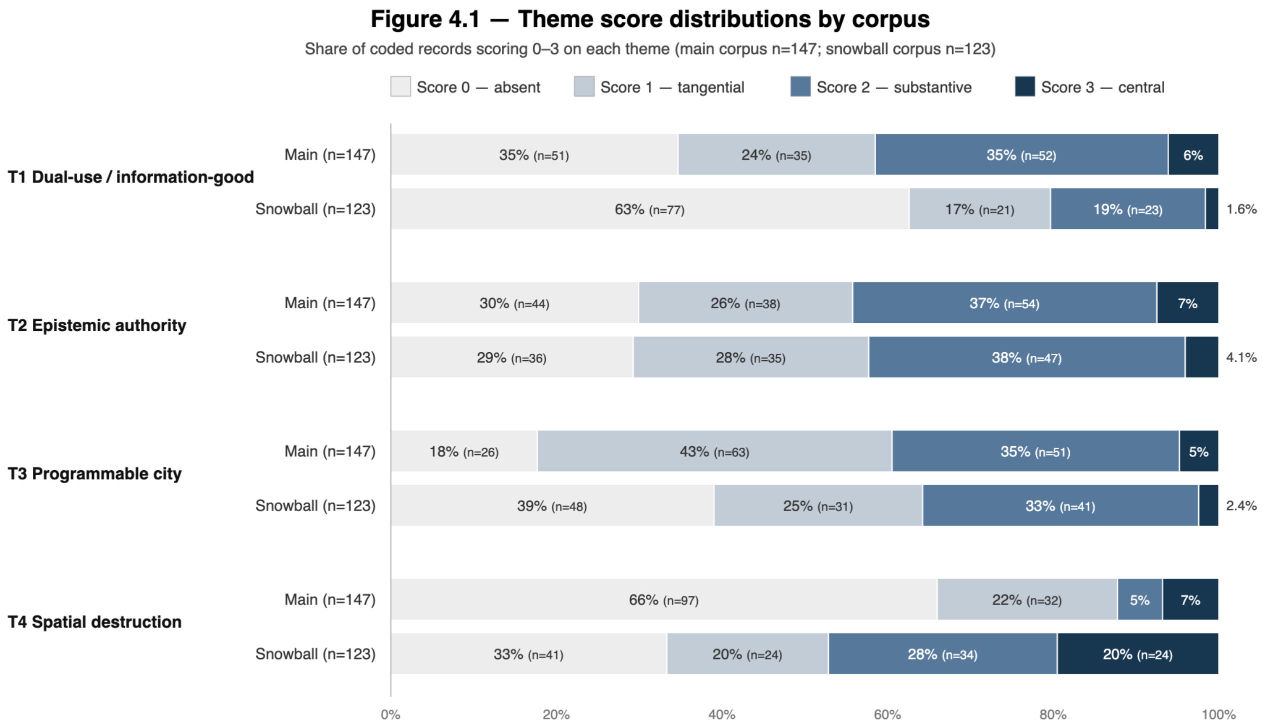

Figure 4.1: Theme score distributions by corpus (share of coded records)

4.2 Theme 1 — The Dual-Use Mechanism ($n = 86$ scoring $\geq$2)

The first theme asked whether the literature treats algorithmic and urban-analytic systems as dual-use: transferable between civilian governance and military application with minimal modification. The answer is yes — but with a structure, and a hole.

4.2.1 Dual-use as design goal: the practitioner literature

The largest cluster (∼18 records) is one this dissertation did not expect to find in such volume: technical and defence-planning literatures that affirmatively engineer the military exploitation of civilian smart-city infrastructure. A series of NATO-adjacent papers proposes architectures for integrating smart-city IoT into military information systems — mapping urban sensor data onto NATO data models for 'situational awareness' in urban operations (Pradhan, 2018; Riberto et al., 2018; Johnsen et al., 2018; Suri et al., 2018). Tsiolis and Leivadeas (2025, p. 18) make the logic explicit in their title: an "Internet of Military Defense Things" that assumes "future military forces will have to conduct a range of operations" inside "permissive" intelligent urban operational environments. The UK DCDC Future Cities reports (Development, Concepts and Doctrine Centre, 2019; Development, Concepts and Doctrine Centre, 2020) analyse smart-city infrastructure as both opportunity and vulnerability for British forces; US Army Engineer Research and Development Center (ERDC) literature advocates "rapid dual-use development of commercial technologies" for force protection on domestic installations — the civilian smart-city stack treated as a military resource.

Two things follow. First, the transfer mechanism the framework predicts (Chapter 2) is not speculative: practitioners name it — interoperability standards (MQTT, NGVA, MIP), commercial off-the-shelf components, open data services. Second, the direction of transfer in this cluster is overwhelmingly civilian-to-military, and it is celebrated rather than lamented. The critical literature's "boomerang" (Graham, 2012, p. 150) here appears as a feature.

4.2.2 The boomerang: war-zone technology comes home

The mirror cluster ($\sim$16 records) is the critical canon on military-to-civilian transfer. Wiig (2018) traces counterinsurgency tactics and technologies from Iraq and Afghanistan into the Camden Metro Police Department. Wall (2013, p. 4) names the "green-to-blue pipeline" by which military drones become domestic policing tools, and explicitly theorises "dual-use scopic technologies" (p. 2). The genealogy of predictive policing runs through the Pentagon: PredPol descends from Army Research Office funding and battlefield-casualty modelling in Iraq (Sharma and Nijjar, 2024; Brayne, 2020), and Palantir's software moved from counterinsurgency to the LAPD (Brayne, 2020). Coaffee and Fussey (2015, p. 95) documents ANPR cameras "developed from former military technology" redeployed across British civic space; Kitchen and Rygiel (2014) track the Long-Range Acoustic Device (LRAD) — a directed-energy weapon developed for maritime force protection — from Somali pirate deterrence to crowd control at the Toronto G20. Schimmel (2021) extends the boomerang logic into the domain of spectacle: the NFL's security infrastructure is legally classified alongside Pentagon suppliers under the SAFETY Act, and stadiums are operationalised as domestic security laboratories — crowd-monitoring technologies tested on football fans migrate to policing contexts.

4.2.3 Commercial data, state security

A third cluster ($\sim$12 records) documents the flow of commercially generated data into security apparatus — the surveillance-capitalism/state nexus. Amoore (2009, p. 50) provides the corpus's most developed account: association-rule algorithms honed on Marks & Spencer and Wal-Mart customer data reappear in US VISIT and e-borders, security practices "oscillating" between commercial and military spheres. Wilhelmsen (2022) shows commercial location data repurposed for espionage; Big Brother Watch (2019) traces Experian's Mosaic marketing segments into Durham Constabulary's Harm Assessment Risk Tool (HART) for recidivism prediction.

4.2.4 Dual-use as geostrategy

A fourth cluster ($\sim$8 records) treats dual-use as state strategy: China's 'safe city' exports as vehicles of civil-military fusion and geopolitical influence (Weber, 2023) — the urban "city brain" whose architecture derives from, and feeds, military command brains (Atha et al., 2020, p. 18); (Robinson, 2020) — with parallel dynamics in Gulf smart-city procurement (Ziadah, 2021).

4.2.5 The hole: nobody prices the transfer

The gap can now be stated precisely. Across 86 records engaging dual-use, the recurring coding annotation — 'does not engage with the economic properties of information goods' — appears on the large majority. The literature documents that transfer happens, richly, in both directions, across dozens of cases. It almost never asks *why* it is frictionless: why software, unlike the bulldozer or the water network, moves between domains at near-zero marginal cost, carrying its ontologies with it. Only Amoore (2009), Wilhelmsen (2022), and Srivastava (2021) gesture at the cost structure of information goods. A distinct older literature on physical dual-use — infrastructure weaponised or sanctioned (Salamanca, 2011; Coward, 2009; Zeitoun, 2018; Coyles, 2017) — makes the contrast visible: pipes and generators can be repurposed only slowly, visibly, and at cost, and dual-use sanctions can bite (Zeitoun's Basrah, where import restrictions on 'dual-use' parts crippled water repair). Software escapes this friction. The corpus thereby validates the dissertation's first theoretical move — Arrow's information-good economics as the missing mechanism — precisely by leaving it unmade.

4.3 Theme 2 — Epistemic Authority and the Black Box ($n = 117$ scoring $\geq$2)

The second theme is the corpus's most populated: how algorithmic systems acquire the authority to define truth, and how that authority is insulated from challenge. Three findings stand out.

4.3.1 The pattern is political

The predictive-policing literature ($\sim$22 records) has matured into a developed epistemology of algorithmic governance. Kaufmann (2019, pp. 675, 690) states the core problem cleanly: patterns are "the epistemological core of predictive policing," and "the clean surface of the pattern makes it impossible to defend oneself against their results." Studies of specific systems show political choices being laundered into code: Rio's CrimeRadar naturalises its designers' assumptions "in the process of translating crime events into strings of codes, datasets, and maps" (Edler Duarte, 2021, p. 203); the pattern's authority is rearticulated with each technical generation (Hälterlein, 2021). Ethnographic work complicates the automation story from inside: Bogotá police "corrupt" crime data at input through institutional cultures (Barreneche, 2019, p. 347); LAPD officers dismiss risk scores as "witchcraft" (Brayne, 2020, p. 87) while nonetheless being governed by them.

4.3.2 Opacity is manufactured, not inherited

The second finding reframes the black box. Across legal and policy records ($\sim$15), opacity is not a technical property of machine learning but an institutional product: trade secrecy asserted by vendors (Brayne, 2020), procurement frameworks where "opacity is normalised as a feature and not a bug" (ARTICLE 19 and Digital Rights Collective, 2022, p. 9), and statutory carve-outs that exempt national security from transparency law — the LED and AI Act "fail to lift the 'veil of secrecy'" (Erdogan, 2024, p. 22), whose individualised remedies "cannot correspond to effects of AI technologies at a population-level." EPIC's amicus work (Electronic Privacy Information Center, 2026, p. 24) catches the resulting bind: automated license plate recognition (ALPR) decisions are "cloaked in objectivity and hidden in a proprietary black box," making constitutional oversight structurally impossible; Erdogan (2024) extends the argument to the European regulatory framework. The implication for the framework: technical transparency fixes (explainable AI, audits) recur in the corpus as necessary but radically insufficient, because the box is legal before it is computational.

4.3.3 Race is constitutive, not contaminating

The third finding carries the corpus's strongest normative charge ($\sim$14 records). Against the 'bias' framing — neutral systems corrupted by skewed data — Sharma and Nijjar (2024, p. 279), quoting Scannell (2019), argue that "policing does not have a 'racist history'. Policing makes race," and algorithms "cannot 'code out' race from American policing because race is an originary policing technology." Ziadah (2021, p. 4) documents "digital epidermalization" in the UAE; Irungbam (2026, p. 11) frames Manipur's internet shutdowns as "epistemicide" — the erasure of subaltern knowledge as a governing technique. The occupation literature extends this to epistemic sovereignty over territory: at Qalandia, "uncertainty becomes the ultimate system of control" (Tawil-Souri, 2011, p. 8); in East Jerusalem the state manages "the uneven distribution of visual rights" (Shalhoub-Kevorkian, 2017, p. 5). This cluster is where Michael's (2007) epistemic-authority concept gets its sharpest empirical elaboration: the authority to define truth is inseparable from the authority to decide whom truth targets.

A bridging motif recurs across all three: the view from above as an epistemic position —

the "God's Eye View" (Lyon, 2014), the drone's "cosmic view" (Wall, 2011, p. 241), the helicopter's "verticalized omniscience" (Adey, 2010, p. 54), Graham's (2012, p. 143) machinic "Id-ing" replacing social identification. Epistemic authority in this literature is literalised as altitude. Theme 4 shows what that altitude is for.

4.4 Theme 3 — The City as Programmable System ($n = 102$ scoring $\geq$2)

The third theme coded whether the literature conceptualises the city itself as a programmable, data-driven system subject to continuous modulation — the Deleuzian-DeLandan ontology (Chapter 2). The answer comes from three directions that rarely cite each other.

Smart-city advocates and their critics share an ontology. Urban Operating Systems are analysed as a "diagram of control" that "does not function to represent… but rather constructs a real that is to come" (Marvin and Luque-Ayala, 2017, p. 7). Realtime urbanism names latency itself as a governing logic that "naturalize[s] control as responsiveness, and collapse[s] deliberation into automation" (Lotfi-Jam, 2025, p. 159). The smart-city critique — "smartmentality" (Kitchin, 2018, p. 224), "algorithmic regulation"[1](Tenney, 2016), "programmable environments" (Williamson, 2015) — takes the same city-as-system ontology as given, contesting its politics rather than its feasibility.

Modulation runs through more channels than dashboards. A distinctive sub-current tracks control exercised through environment and affect: "scripted architecture" using light, smell, and sound as "pastoral power" (Schuilenburg and Peeters, 2018, p. 7); the Eindhoven nightlife living lab whose operators are "Big Brother only to the Dialogue masses" (Jacobs and Van Ostaijen, 2024); "atmospheric fortification" (Fregonese, 2024, p. 1); time itself as "an infrastructure of control" in East Jerusalem (Baumann, 2019, p. 4).

The military independently arrives at the same city. The hinge finding of this theme: defence doctrine converges on the identical ontology. DARPA's Combat Zones That See aimed to build "fully representative data profiles on the 'normal' time-space movement pat-

[1] On Volunteered Geographic Information (VGI), Tenney (2016, p. 106) notes the self-reinforcing dynamic whereby systems that generate ever-larger data volumes require further algorithmic systems to govern them.

terns of entire subject cities” so that algorithms could flag the abnormal (Graham, 2008, p. 35). US military urban doctrine treats the city as a “system-of-systems,” a “living organism” to be managed (Danielsson, 2025, pp. 406, 413); China’s “city brain” is explicitly an OODA[2] loop(Weber, 2023, pp. 3–4). And a corrective to presentism runs underneath: the administered and occupied city modulated populations bureaucratically long before software — Belfast’s “controlled allocation” of tenancies as pre-emptive spatial design (Coyles, 2017, p. 9), the French colonial Section Administrative Urbaine’s (SAU) remapping of Algiers “into militarized zones” (Crane, 2017, p. 198), Gaza’s electricity and water switched on and off as “elastic” humanitarian calibration (Salamanca, 2011, p. 1). Algorithmic governance automates and intensifies an older administrative modulation; it does not invent it. The programmable-city ontology, then, is not contested between the literatures — it is shared by smart-city advocates, critical urban theorists, and military doctrine alike; what is contested is only who holds the controls.

4.5 Theme 4 — Spatial Destruction and Home Unmaking ($n = 76$ scoring $\geq$2)

The fourth theme is the corpus’s most intense: 34 of 76 records score 3, the highest saturation of any theme. This is the literature the keyword search missed and the citation network delivered (Table 4.1).

4.5.1 The urbicide canon and its paradigmatic case

The conceptual spine runs from Graham’s (2004) “postmortem city” — war as “the most thorough-going… occasion of collective violence that destroys places” (Hewitt, cited therein) — through urbicide’s tripartite definition: the killing of cosmopolitan mixing, of the means of modern urban living, and of those cast as “unmodern” (Graham, 2012, p. 148). Hanafi (2012, pp. 190, 192) “spacio-cide” names the Palestinian variant: “the weapons of mass destruction

[2] Observe, Orient, Decide, Act: the decision loop developed by US Air Force strategist John Boyd, now widely adopted beyond its military origins.

are not so much tanks as bulldozers.”

Palestine is the corpus's gravitational centre ($\sim$25 records) and functions explicitly as a laboratory (Cook, 2009, p. 7: confined spaces as “laboratories where experiments to encourage Palestinian despair… are being refined”). Every modality of destruction is documented: home demolition as planning instrument (Meade, 2011; Paz-Fuchs, 2010); “walking through walls” (Bleibleh, 2015); closure, checkpoint, and enclave (Peteet, 2015, p. 10) — Weisglass's “diet, but not… die of hunger”; settlement as territorialisation (Volinz, 2019); sensory warfare (Shalhoub-Kevorkian, 2017). Golańska (2022, p. 1) “slow urbicide” names the attritional register: bureaucratic neglect, pollution, obliteration of the vernacular as violence at low visibility. And the register has accelerated: Benguita (2025, p. 399) reads Gaza 2023–25 as “Spatial Nakba” — “Gaza is not merely being bombed — it is being unmade as a space of collective identity and political potential” (p. 411) — while Holail et al. (2024) quantify the escalation through time-series satellite sensing.

4.5.2 Destruction as cycle: razing, renewal, razing

A second cluster corrects any event-based reading. Urban renewal produces the displaceability that later warfare exploits (Genç, 2021; Coyles, 2017), and post-conflict reconstruction continues the war by other means: Turkey's tabula rasa rebuilding of Sur replaces “vernacular and communal architecture” with housing that “integrate[s residents] into state agencies and mortgage regimes” (Smith, 2022, p. 16); Hourani (2024, p. 2) names the “post-conflict-catastrophe complex.” Wilson and Wyly (2026, p. 2) push the continuum furthest into peacetime, introducing “Dracula urbanism” to name the process by which real-estate development itself becomes a mode of slow domicide: displacement by market logic, the violence of eviction made administratively routine. Rio's “Museum of Evictions” (Arantes and Ribeiro, 2022, p. 387), Cape Town's securitised renewal (Samara, 2010), and the MOVE bombing's destruction of a Philadelphia block (Massaro, 2015) extend the continuum across regime types. Domicide, the corpus insists, is a spectrum practice of states, not a wartime aberration.

4.5.3 The warrant: dual-use inside the destruction

The single most important finding for the dissertation's thesis appears here. Benguita's (2025) coding of Gaza 2023–25 surfaces the exact vocabulary: "Hospitals, schools, mosques, media institutions, and universities have been struck under the rationale of *dual-use* or symbolic alignment with resistance" (p. 406) (emphasis added). The legal-military category of *dual-use* — the same category that governs export controls and sanctions in Theme 1 — operates inside the destruction literature as the warrant for striking civilian fabric. Coward (2009, p. 406) documented the same logic in Iraq's "dual use facilities"; Graham (2011, p. 355) recorded the USAF planner's concession that targeting dual-use electrical infrastructure killed up to 100,000 civilians. The thesis that algorithmic governance and spatial destruction share a dual-use logic is therefore not an analogy this dissertation imposes on the material: it is a connection the material states about itself.

4.5.4 The thin edge: algorithmic destruction

Yet the algorithmic mode of destruction is the theme's thinnest cluster. Five records in the coded corpus of 270 substantively address algorithmically mediated spatial destruction — four published since 2021, the fifth a precursor from 2008. Because these five are the review's most direct evidence at the junction the framework predicts, each is taken in turn.

The precursor is Graham's (2008) reading of DARPA's Combat Zones That See programme as the blueprint for the algorithmically targeted city. The programme aimed to saturate global-South cities with networked surveillance systems, building "fully representative data profiles on the 'normal' time-space movement patterns of entire subject cities" (p. 35) so that algorithms could flag the abnormal. Written before the smart city had a name, it is the corpus's earliest statement that the targetable city and the optimised city require the same infrastructure: continuous, population-scale, machine-readable modelling of urban life.

Where Graham supplies the blueprint, the Završnik and Badalič volume *Automating Crime Prevention, Surveillance, and Military Operations* (Završnik and Badalič, 2021) — above all Hogue's (2021, p. 203) dissection of Project Maven, the Pentagon's flagship computer-vision programme — supplies the epistemic mechanism: automation changes knowledge production itself, since "Algorithmic vision produces a reality from which to act." Applied to targeting, this relocates destruction upstream: the kill chain begins not with the strike but with the categories — person of interest, pattern of life — that make targets thinkable; the black box is not adjacent to the violence but its first stage.

The kill chain itself arrives with Hourani's (2024, p. 3) account of war, reconstruction, and the city — the corpus's most direct documentation of algorithmically mediated killing: Israel's Lavender and Where's Daddy? systems generate targets at scale and track them "to their homes, where they are killed along with their families," a practice that "gives lie to technophilic dreams of the sanitary smart bomb." Hourani situates these systems within the post-conflict-catastrophe complex, the cyclical logic (§4.5) by which destruction and reconstruction continue the same war, and warns they "may be indicative of what's to come": mass-produced targeting as standing capability rather than emergency measure.

The fourth record works in the opposite direction, as a counter-proof. Holail and colleagues (2024, p. 2) pair LuoJia3-01 satellite imagery with deep-learning damage detection to quantify Gaza's destruction in near-real time: 15.4% of structures destroyed, 18.7% severely damaged, and over 3,700 missile craters mapped across five governorates — the only quantitative record of the five. It both defines and measures "urban annihilation" as a detectable transformation of the built environment, and demonstrates the sensing stack's documentary afterlife: the same class of infrastructure used for targeting can be turned to accountability — the counter-forensic possibility taken up in the discussion (§5.4).

The fifth, Lotfi-Jam's (2025) "realtime urbanism," is the only one of the five to engage all four framework themes, and reads as their synthesis. Its argument: the capacity "to sense, model, and intervene with minimal delay" (p. 160) is a single operational condition shared by military, entertainment, and urban systems, in which latency becomes a governing logic that "naturalize[s] control as responsiveness, and collapse[s] deliberation into automation" (p. 159). Read against the framework, this is Arrow's frictionlessness and Deleuze's modulation restated at the level of infrastructure: when sensing and intervention converge to near-zero delay, the distinction between managing a city and striking it becomes a parameter setting — and the AI-assisted kill chain is the proof.

The provenance of these five records vindicates the review's dual-instrument design (§3.3). Two arrived through Search A's governance vocabulary (Završnik and Badalič, 2021; Lotfi-Jam, 2025), two through Search B's destruction vocabulary (Graham, 2008; Holail et al., 2024), and one through forward chaining from Graham (2006) (Hourani, 2024). No single instrument recovered more than two of the five, and each recovered records invisible to the others: the junction's thinness is not an artefact of any one search strategy.

Against 76 destruction records and 117 epistemic-authority records, these five constitute the only sustained bridges. Four of the five are theoretical; four date from 2021 or later; and the theoretical four lean heavily on investigative journalism for their operational detail. The urbicide canon theorises destruction but barely touches its algorithmisation; the algorithmic-governance literature theorises modulation but stops short of the kill chain. The intersection — algorithmically mediated urbicide — is nearly empty. This is the dissertation's central empirical finding, and it is a finding about the literature itself.

4.6 The Map as Argument

Read together, the four themes form the causal chain the framework predicted, and the corpus maps each link with sharply varying density:

- **Mechanism (T1, 86):** transfer is documented exhaustively, in both directions — but its economic grammar is untheorised.
- **Authority (T2, 117):** the black box is shown to be institutionally manufactured, and its truths constitutively racialised.
- **Ontology (T3, 102):** the programmable city is shared by smart-city advocates, critical theorists, and defence planners — a common operational picture whose politics depend entirely on the institutional context of deployment.
- **Consequence (T4, 76):** destruction is theorised intensely (34 records at score 3) — but its algorithmic mediation is documented in only five records.

Two further patterns sharpen the picture. Direction of transfer is asymmetric: among records engaging transfer explicitly, civilian-to-military dominates (n=65) over military-to-civilian (n=20) and bidirectional (n=32) — the pipeline runs outward from the smart city more often than it returns. And structural dual-use — the system-level property the dissertation's definition requires, beyond individual artefacts — is explicit in only 37 of 270 coded records. The literature is rich on cases, thin on structure.

The corpus's own architecture performs the thesis one final time. The governance literature and the destruction literature were found by different instruments, cite different canons, and score inversely on the framework's first and last themes (Table 4.1). They are two conversations about one assemblage, conducted in separate rooms.

To state the chapter's outcome plainly. The four themes are confirmed as the literature's organising concerns, but they are unequally provisioned: the mechanism and the authority are densely mapped, while the junction between them and the destruction they enable is nearly empty. And the single most consequential connection — dual-use operating as a warrant for destruction — appeared unbidden, inside the destruction literature's own vocabulary. The map thus confirms the framework's elements while refuting its assumed connectedness. What that refutation means — for the dual-use claim, for policy, and for a research agenda on algorithmically mediated domicide — is the work of Chapter 5.

Chapter 5

Discussion

5.1 The Claim, Tested

Chapter 2 staked the dissertation on a specific claim: that algorithmic governance tools are structurally dual-use — transferable between civilian urban management and military spatial destruction with a fluidity that physical infrastructure cannot match — and that the framework locates the sources of this fluidity in the economics of software, the institutional logic of control societies, and the epistemic authority of algorithmic systems. The systematic map in Chapter 4 lets that claim be tested against 270 coded records rather than asserted from a handful of canonical texts. The verdict is more nuanced than simple confirmation. The literature supports each element of the framework — but it supports them in different literatures, and the junction between them is nearly empty.

5.1.1 Arrow: present in the phenomenon, absent in the explanation

The economic mechanism — Arrow's near-zero marginal cost of reproduction as the substrate of dual-use fluidity — is the framework's essential element and the literature's loudest silence (§4.2). Eighty-six records document transfer between civilian and military domains; almost none theorise its cost structure. The physical-dual-use counterexamples make this absence instructive: in Zeitoun (2018)'s Basrah, dual-use sanctions bit on physical assets — steel tankers, spare parts, the material supply chains of state infrastructure. In Salamanca

(2011, p. 8)'s Gaza, weaponised electricity required switches, fuel lines, and an ongoing military siege. Software escapes all this friction. The corpus — practitioner papers designing the transfer (Tsiolis and Leivadeas, 2025) no less than critical accounts deploring it (Amoore, 2009; Wall, 2013) — simply takes that escape for granted. The Arrow-based mechanism is thus untested but consistent with everything the corpus contains: the phenomenon is documented pervasively, its explanation is absent, and the framework's explanation contradicts nothing in the record — suggestive rather than confirmatory, it marks the cost structure of dual-use transfer as a priority for dedicated empirical research, not an established result.

5.1.2 Deleuze and DeLanda: confirmed by convergence

The control-society and machinic-assemblage elements fare differently: they are not absent but saturated (§4.4). Deleuze travels explicitly across policing, smart-city, and occupation cases alike. More telling is the convergence the review was designed to detect: military doctrine — DARPA's Combat Zones That See profiling the "normal" rhythms of "entire subject cities" (Graham, 2008, p. 35), the "military urban" as system-of-systems (Danielsson, 2025, p. 406), the city brain as OODA loop (Weber, 2023, pp. 3–4) — arrives independently at precisely the ontology that DeLanda's targetable city predicts and that smart-city practitioners advertise. When smart-city advocates, critical theorists, and defence planners share an ontology, the ontology is no longer a critical imposition; it is a description. And the analogue cases (Coyles, 2017; Crane, 2017) dissolve the obvious objection — that the targetable city is hype — by showing the same modulation logic operating bureaucratically decades before the dashboards. Software automates it.

5.1.3 Michael: confirmed and extended

The epistemic-authority element is the corpus's most populated theme (n=117), and the literature extends Michael in two directions he did not go. First, institutional: the black box is manufactured — by trade secrecy, procurement frameworks, and statutory carve-outs (§4.3) — which relocates epistemic authority from the general's briefing room to the vendor contract and the regulation's exemption clause. Second, constitutive: the strongest records reject the ‘bias’ frame in favour of race as an originary property of what these systems are built to do (Sharma and Nijjar, 2024; Ziadah, 2021). Michael’s claim was that the military became the arbiter of truth when civilian politics abdicated. The corpus suggests the algorithmic version is more radical: the system does not merely arbitrate truth, it produces the categories — ‘person of interest,’ ‘pattern of life,’ ‘associated’ — within which truth-claims are thinkable at all (Amoore, 2009).

5.1.4 King, Graham, Weizman: the canon holds, the link is missing

The destruction canon emerges intact and intensely alive — 34 of 76 Theme-4 records score 3 — and is currently undergoing its fastest expansion, driven by Gaza 2023–25 (Benguita, 2025; Holail et al., 2024). Its key internal development is the cyclical reading: renewal produces displaceability, razing follows, reconstruction continues the war by other means (Genç, 2021; Smith, 2022; Hourani, 2024), with “Dracula urbanism” (Wilson and Wyly, 2026, p. 2) extending domicide into peacetime development logic. What the canon does not yet contain is the algorithmic mode of destruction. That absence is the subject of the next section.

5.2 The Empty Intersection

The review’s central empirical finding is negative: the intersection of the algorithmic-governance literature and the spatial-destruction literature — records that substantively treat algorithmically mediated urbicide — comprises five records (§4.5.4), all but one published since 2021. The two literatures were recovered by different instruments and score inversely on the framework’s first and last themes (Table 4.1); their citation canons barely touch. Even the snowball’s anchor structure tells the story: Kitchin’s network produced the governance side, Graham’s and Weizman’s the destruction side, and Michael’s — the theorist whose concept bridges them — produced nothing the other networks had not already captured.

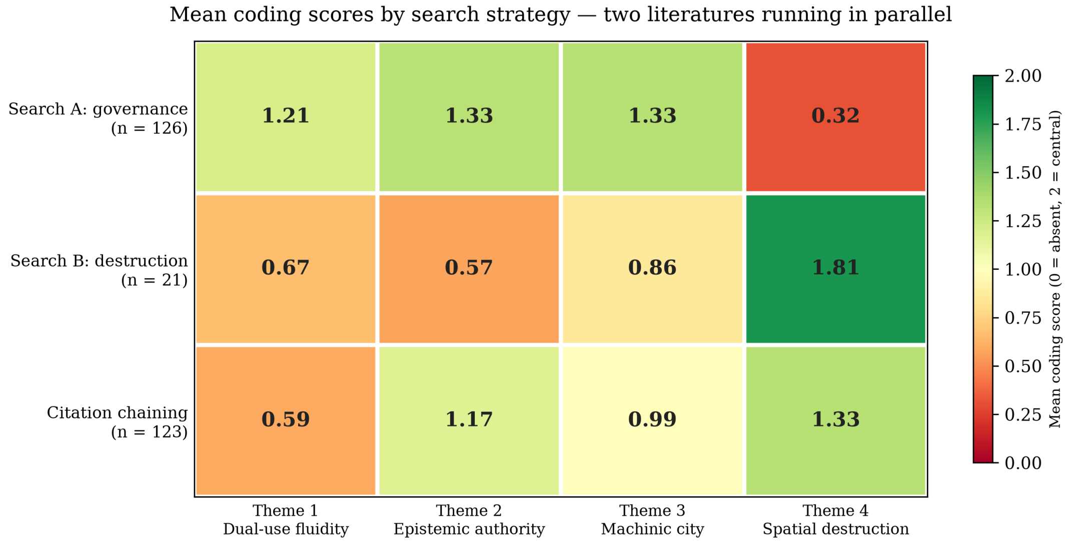

Figure 5.1: Mean coding scores by search strategy across the framework's four themes. The governance literature (Search A) saturates Themes 1–3 but is nearly silent on spatial destruction; the destruction literature (Search B) shows the inverse pattern; citation chaining deepens both sides without bridging them.

Three explanations are plausible, and the corpus cannot fully adjudicate between them. A zero possibility must first be discounted: that the emptiness is an artefact of the search design. Search A was built for high recall of governance literature, Search B for high precision on destruction literature; two deliberately non-overlapping instruments returning non-overlapping literatures would, on its own, prove little. Two features of the review resist that reading. The citation chaining was search-term-independent, yet it reproduced the same separation — and the 124 chained records *deepened* both sides without bridging them. And the asymmetry of the five bridge records runs one way: destruction-side authors reaching for algorithmic governance, never governance-side authors reckoning with destruction (§4.5.4). The intersection is empty in the literature, not merely in the search.

Of the substantive explanations, the first is disciplinary path dependence: urbicide theory descends from critical geography and architecture; algorithmic-governance critique from surveillance studies and STS; their conferences, journals, and citation graphs barely touch. The second is secrecy: the systems that would constitute the intersection — Lavender, Where's Daddy?, the Gospel — are classified and contested; scholars can access them only through investigative journalism (Hourani, 2024), which throttles the normal machinery of theoretical elaboration. The third is timing: the large-scale deployment of

AI target-generation in an urban war is, at the time of writing, roughly two years old. The literature is not absent; it is embryonic. On the third reading, the Gaza case carries a double status in this dissertation: paradigmatic for the warrant logic documented in §5.3 — which has a decades-long documentary record — but, in its algorithmic dimension, an event the literature has barely begun to register; there the framework's claims are predictive rather than explanatory. The empty intersection is then not evidence against the framework but evidence of the framework's timeliness — and it fixes the dissertation's contribution as cartographic: naming a junction before the traffic arrives.

5.3 Dual-Use as Warrant

One finding exceeded the framework, inverting the usual direction of critique. The framework treats dual-use as an analytic category — a lens for seeing how algorithmic systems travel between domains. But the corpus shows dual-use operating as a juridical category inside the machinery of destruction: hospitals, schools, mosques, and universities "struck under the rationale of dual-use" (Benguita, 2025, p. 406); "dual use facilities" as the sanction-era targeting warrant in Iraq (Coward, 2009, p. 406); dual-use electrical infrastructure as the targeting category whose deliberate degradation cost up to 100,000 civilian lives (Graham, 2011, p. 355). The Iraqi power grid was not destroyed incidentally; it was targeted specifically because its dual-use character — electricity serves both civilian hospitals and air-defence radars — made it militarily licit under prevailing interpretations of IHL. Gaza 2023–25 operates the same warrant on a vastly expanded scale, where the civilian data infrastructure itself — population maps, building footprints, connectivity graphs — becomes both the targeting instrument and the targeting justification. The category that export-control law uses to restrain the flow of sensitive technology is the same category targeting doctrine uses to license the destruction of civilian fabric. Dual-use names the collapse of the civilian/military distinction — and that collapse is as much the targeter's resource as the critic's concern. Regulating the category's supply side while leaving its targeting function intact polices the flow of a logic it does not touch.

The warrant logic is not confined to targeting doctrine; it is stated, by at least one state, as national strategy. Israel's *National Initiative for Secured Intelligent Systems* — the 2020 Special Report to the Prime Minister — explicitly frames intelligent systems as a strategic technology whose national value lies in the dual combination of security needs and economic opportunity, and places that duality at the centre of state planning (Ben-Israel et al., 2020). This is the dual-use logic in the language of national doctrine, from the state whose military would, within three years, be deploying AI-driven target-generation systems in an urban war. The convergence of strategic policy discourse and operational military practice is not coincidental: it is the structural dual-use of this dissertation's framework, formalised at the highest level of state planning.

5.4 Policy Implications

For a public-policy dissertation, the map carries four implications, escalating in specificity. A caveat first: the corpus documents what the literature says; it does not test regulatory alternatives. What follows is the author's policy-interpretive extension of the map, and a structural analysis of this kind cannot by itself settle questions of instrument design — alternative readings of the same evidence are possible, and the implications below are offered as the reading the framework most strongly supports.

Agency, however, is not dissolved by structure. The framework describes conditions that make certain transfers easy and certain oversights likely; it does not make them automatic. Vendors design ontologies, consultants recommend platforms, procurement boards sign contracts, ministers approve carve-outs, targeting officers accept machine-generated lists. The structural account locates where those decisions cluster and why they currently escape scrutiny — it is an argument about where to look for responsibility, not a reason to stop looking.

First, *regulate the ontology, not the artefact*. Current dual-use regimes enumerate items — sensors, chips, software categories. The corpus shows transfer moving through data models, ontologies, and interoperability standards (§4.2), and through vendor contracts that carry military logics into civilian administration wholesale (Palantir's counterinsurgency-to-LAPD pipeline; the HART tool built on Experian marketing segments). A governance regime that licences artefacts while leaving the ontological layer unexamined will be perpetually one step behind the transfer it purports to control.

Second, *opacity is a policy output and can be a policy target*. If the black box is manufactured by procurement and statute (§5.1), it can be unmade by them: transparency conditions written into public procurement of algorithmic systems, population-level impact assessment rather than individualised remedies (Erdogan, 2024), and statutory refusal of national-security carve-outs for municipal deployments. These are modest, actionable instruments — and the corpus suggests they address the box where it actually lives.

Third, *treat 'smart city' procurement as security policy*. The practitioner literature already does (§4.2): NATO-adjacent papers design the military use of civilian urban IoT; defence assessments treat adversary smart-city exports as strategic threats (Atha et al., 2020; Weber, 2023). Municipalities buying integrated urban platforms are, on the evidence of this corpus, procuring battlespace infrastructure — whether or not they intend to. Democratic oversight that begins at the police procurement stage arrives years too late; it must begin at the platform contract.

Fourth, *the warrant problem requires attention to the targeting side of dual-use*. Section 5.3 implies that humanitarian-law advocacy cannot cede the dual-use category to military lawyers. The emerging counter-forensic practice — using the same sensing stacks to document destruction that targeting uses to produce it (Holail et al., 2024) — is the corpus's clearest existing model of contestation at this level, and it deserves policy support: open damage-assessment data, funded documentation mandates, and evidentiary standards for algorithmically generated targets.

These implications assume a regulatory register. A stronger position should be named explicitly: that certain classes of algorithmic urban technology should be prohibited rather than regulated, on the grounds that the structural dual-use property this review documents makes safe regulation impossible. That position is coherent and consistent with this evidence; this dissertation does not advocate it, but the evidence does not refute it either.

Two wider currents in policy studies converge on these implications. The first is the evidence-and-evaluation debate: four decades of the 'what works' turn (Tranfield et al., 2003) have built sophisticated machinery for evaluating policy interventions, while the algorithmic systems that increasingly *execute* those interventions have escaped equivalent scrutiny; the first two implications above would extend the evaluative apparatus to the executants. The second is the recognition — from infrastructure studies to platform regulation — that governance increasingly happens at the layer of technical standards and data models rather than statutes; the third and fourth implications apply that recognition to the dual-use case.

5.5 What the Map Cannot Say

Two constraints bear directly on everything above. First, the review maps public discourse: the classified systems that would most directly test the thesis are absent by definition, and the five-record intersection may partly reflect that absence rather than a true gap in the world (§3.6). Second, the map cannot adjudicate prevalence: because the search strategies were designed to recover dual-use and destruction literature rather than to sample urban-technology discourse representatively, the corpus cannot say how typical dual-use transfer is within the wider field of civilian urban analytics, and it offers no measure of the civilian benefits those systems produce. The conclusions in Chapter 6 are calibrated accordingly: the dissertation claims to have mapped and explained a discursive junction, not to have demonstrated the operational reality of any specific targeting system.

Chapter 6

Conclusion and Research Agenda

6.1 What the Dissertation Set Out to Do

This dissertation began with an observation and a wager. The observation: the same data infrastructures that promise to optimise the civilian city — its traffic, its hospitals, its policing — are structurally available to the apparatus that destroys cities. The wager: that this is not a metaphor but a mechanism, with identifiable economic, institutional, and epistemic components, and that a systematic review of three and a half decades of literature could test it. Chapter 2 assembled the mechanism from five theoretical elements: Arrow's information-good economics, Deleuze's control societies, DeLanda's targetable city, Michael's epistemic authority, and the urbicide canon of King, Graham, and Weizman. Chapters 3 and 4 built the test: a PRISMA-guided review of 2,713 records screened across two identification streams, yielding a coded corpus of 270, mapped against the framework's four themes.

6.2 What It Found

Three findings stand. First, the framework's elements are each confirmed, but in segregated literatures. Dual-use transfer is documented exhaustively in both directions — yet its economic grammar is untheorised, leaving Arrow's mechanism as the dissertation's first contribution. The programmable-city ontology is common ground for smart-city advocates, their critics, and military doctrine alike. Epistemic authority is shown to be institutionally manufactured and constitutively racialised, extending Michael beyond his original formulation. And the destruction canon is thriving, cyclical, and increasingly quantified.

Second, the intersection is empty. Only five records substantively address algorithmically mediated spatial destruction, all but one since 2021. The literature of how algorithms govern and the literature of how cities are destroyed run in parallel — a gap reproduced in the review's own architecture, where keyword search and citation chaining recovered the two sides independently. The dissertation's central empirical contribution is this map of a junction that has not yet been built.

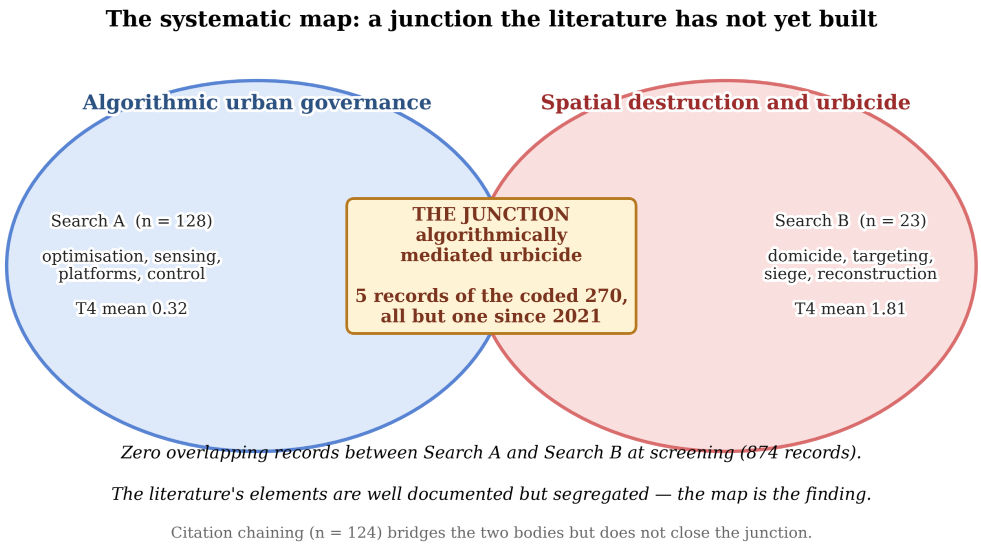

Figure 6.1: The systematic map. Two well-documented literatures run in parallel — algorithmic urban governance (Search A) and spatial destruction (Search B) — with the junction of algorithmically mediated urbicide substantively addressed by only five records of the coded 270, all but one published since 2021.

Third, dual-use operates as a warrant, not merely a flow. The category through which export law restrains technology transfer is the same category through which targeting doctrine licenses strikes on civilian fabric — "hospitals, schools, mosques… struck under the rationale of dual-use" (Benguita, 2025, p. 406). The collapse of the civilian/military distinction is the targeter's resource as much as the critic's concern. This finding, unanticipated in the framework, is the review's sharpest policy-relevant result.

6.3 An Agenda for Research

The empty intersection defines the agenda. Five directions follow.

1. Name the field. Algorithmically mediated urbicide needs its own literature before it can have its own debates. The five-record beachhead (Hourani, 2024; Lotfi-Jam, 2025; Završnik and Badalič, 2021; Graham, 2008; Holail et al., 2024) should be consolidated — through a dedicated special issue or state-of-the-field review — into a citable research programme connecting critical geography, surveillance studies, and International Humanitarian Law (IHL) scholarship.

2. Quantify the junction. The corpus is 7% quantitative. Remote-sensing damage assessment (Holail et al., 2024) shows that the sensing stack's documentary use is feasible; systematic work pairing targeting disclosures with damage signatures could move the field from assertion to measurement.

3. Extend the comparative base. The destruction literature's Palestine concentration is both its strength and its constraint. Kurdish Turkey, Belfast, Rio, Kigali, and Basrah appear in the corpus as comparative footholds; systematic cross-case work would test whether the cyclical model — renewal, razing, reconstruction — generalises, and whether the algorithmic mediation of destruction differs across regime types.

4. Institutionalise counter-forensics. The corpus's clearest existing model of contestation is the counter-forensic use of the same stacks that target (Forensic Architecture; Holail et al., 2024). What mandates, funding lines, and evidentiary standards would make that practice a durable check rather than an activist art?

5. Follow the ontologies into the era of autonomous weapons. The review showed transfer moving through data models and interoperability standards rather than discrete artefacts. This finding connects directly to the urgent field of AI interpretability and alignment: the target ontologies, urban data models, and 'pattern of life' schemas that travel between civilian platforms and military systems encode assumptions about what constitutes a legitimate target, a normal civilian, and an anomalous pattern. As lethal autonomous weapon systems (LAWS) transition from development to deployment, these ontologies become the de facto rules of engagement — machine-readable categories that determine who lives and who dies. Research is needed on the specific ontologies that mediate this transition, on the vendor contracts that carry them into operational contexts, and on the alignment failures they embed. This is empirical, document-based work that public administration is well placed to do, and it is the agenda's most consequential item.

6.4 Closing

The introduction opened with a loop: Palantir's software moving from insurgent-mapping to NHS bed management, and urban-analytic sensing moving from peacetime planning to target generation in Gaza. The systematic map shows that loop is not an anecdote but a structure — documented on both sides, theorised on neither. If the review has a single lesson for policy, it is that the relevant governance question is no longer whether a given urban technology can be turned to destruction; the corpus answers that in the affirmative, eighty-six times over. The question is whether democratic institutions will develop the categories — ontological, legal, evidentiary — to see the turn before it completes. At present, as the five records at the empty centre of this map attest, the literature has barely begun to look. That is the gap this dissertation has mapped, and the work it invites.

Bibliography

Adey, Peter (2010). “Vertical Security in the Megacity: Legibility, Mobility and Aerial Politics”. In: *Theory, Culture & Society* 27.6, pp. 51–67.

Amoore, Louise (2009). “Algorithmic War: Everyday Geographies of the War on Terror”. In: *Antipode* 41.1, pp. 49–69.

Anderson, Chris (2008). “The End of Theory: The Data Deluge Makes the Scientific Method Obsolete”. In: *Wired*. 23 June 2008.

Arantes, Pedro Fiori and Cláudio Rezende Ribeiro (2022). “Western Fantasy and Tropical Nightmare”. In: *The Routledge Handbook of Architecture, Urban Space and Politics, Volume I*. Routledge, pp. 379–394.

Arroub, Ayoub, Bassma Zahi, Essaid Sabir, and M. Sadik (2016). “A Literature Review on Smart Cities: Paradigms, Opportunities and Open Problems”. In: *2016 International Conference on Wireless Networks and Mobile Communications (WINCOM)*. IEEE, pp. 180–186.

Arrow, Kenneth J. (1962). “The Rate and Direction of Inventive Activity: Economic and Social Factors”. In: *Economic Welfare and the Allocation of Resources for Invention*. Princeton University Press, pp. 609–626.

ARTICLE 19 and Digital Rights Collective (2022). *Who Buys and Controls CCTV? Myanmar’s Slippery Slope to Mass Surveillance*.

Atha, Katherine, John Callahan, Jay Chen, Jessica Drun, Keric Green, Brian Lafferty, Joe McReynolds, James Mulvenon, Benjamin Rosen, and Eric Walz (2020). “China’s Smart Cities Development”. In: *U.S.–China Economic and Security Review Commission Research Report*.

Barreneche, Carlos (2019). “Governing by Geosurveillance: Predictive Policing and the Question of Space”. In: *Surveillance & Society* 17.3/4, pp. 350–367.

Baumann, Hanna (2019). “Disrupting Movements, Synchronising Schedules: Time as an Infrastructure of Control in East Jerusalem”. In: *City* 23.1, pp. 72–88.

Ben-Israel, Isaac, Eviatar Matania, and Leehe Friedman (2020). *The National Initiative for Secured Intelligent Systems to Empower the National Security and Techno-Scientific Resilience: A National Strategy for Israel*. Special Report to the Prime Minister. In Hebrew. Tel Aviv: Yuval Ne’eman Workshop for Science, Technology and Security, Tel Aviv University.

Benguita, Mourad (2025). “Urbicide in Israel’s War on Gaza: Architecture of Erasure and the Unmaking of Urban Life (2023–2025)”. In: *Filistin Araştırmaları Dergisi (Bulletin of Palestine Studies)* 18, pp. 397–417.

Big Brother Watch (2019). *Written Evidence to the Joint Committee on Human Rights on The Right to Privacy (Article 8) and the Digital Revolution Inquiry*.

Bleibleh, Sahera (2015). “Walking Through Walls: The Invisible War”. In: *Space and Culture* 18.2, pp. 156–170.

Brayne, Sarah (2020). *Predict and Surveil: Data, Discretion, and the Future of Policing*. New York: Oxford University Press.

Bresnahan, Timothy F. and Manuel Trajtenberg (1995). “General Purpose Technologies: ‘Engines of Growth’?” In: *Journal of Econometrics* 65.1, pp. 83–108.

Coaffee, Jon and Pete Fussey (2015). “Constructing Resilience Through Security and Surveillance: The Politics, Practices and Tensions of Security-Driven Resilience”. In: *Security Dialogue* 46.1, pp. 86–105.

Cook, Jonathan (2009). *Disappearing Palestine: Israel’s Experiments in Human Despair*. Bloomsbury Publishing Plc.

Coward, Martin (2009). “Network-Centric Violence, Critical Infrastructure and the Urbanization of Security”. In: *Security Dialogue* 40.4–5, pp. 399–418.

Coyles, David (2017). “The Security-Threat-Community”. In: *City* 21.6, pp. 730–750.

Crane, Sheila (2017). “The Casbah City: The Spatial Production of Insurgency and Counter-Insurgency in Algiers”. In: *City* 21.3–4, pp. 348–364.

Cugurullo, Federico, Federico Caprotti, Matthew Cook, Andrew Karvonen, Pauline McGuirk, and Simon Marvin (2024). “The Rise of AI Urbanism in Post-Smart Cities: A Critical Commentary on Urban Artificial Intelligence”. In: *Urban Studies* 61.6, pp. 1168–1182.

Danielsson, Anna (2025). “The Emergence of a Military Urban in and of War”. In: *Annals of the American Association of Geographers* 115.2, pp. 404–418.

DeLanda, Manuel (1992). *War in the Age of Intelligent Machines*. New York: Zone Books.

— (2014). *Intensive Science and Virtual Philosophy*. London: Bloomsbury.

Deleuze, Gilles (1990). “Postscript on the Societies of Control”. In: *October* 59, pp. 3–7.

Development, Concepts and Doctrine Centre (2019). *Future Cities: Trends and Implications for Urban Living and Military Operations*. UK Ministry of Defence.

— (2020). *Future Cities: Trends and Implications for Urban Living and Military Operations*. UK Ministry of Defence.

Dijck, José van (2014). “Datafication, Dataism and Dataveillance: Big Data Between Scientific Paradigm and Ideology”. In: *Surveillance & Society* 12.2, pp. 197–208.

Edler Duarte, Daniel (2021). “The Making of Crime Predictions: Sociotechnical Assemblages and the Controversies of Governing Future Crime”. In: *Surveillance & Society* 19.2, pp. 199–215.

Electronic Privacy Information Center (2026). *Brief of* Amicus Curiae *Electronic Privacy Information Center*. Supreme Court of the United States.

Erdogan, Ebru (2024). “Diving into the Iceberg: Establishing Transparency in AI for Law Enforcement”. In: *European Journal of Law and Technology* 15.1.

Forensic Architecture (2023). *A Cartography of Genocide: Israel’s Conduct in Gaza Since October 2023*. `https://forensic-architecture.org`.

Foucault, Michel (1975). *Surveiller et punir: Naissance de la prison*. Paris: Gallimard.

Fregonese, Sara (2024). “Atmospheric Geographies of (Counter)Terrorism”. In: *Progress in Human Geography* 48.1, pp. 3–24.

Genç, Fırat (2021). “The Politics of Urban Warfare: Urban Transformation, Violence and Resistance in Diyarbakır”. In: *Political Geography* 89, p. 102445.

Golańska, Dorota (2022). “Slow Urbicide: Accounting for the Shifting Temporalities of Political Violence in Palestine”. In: *Time and Temporality in Transitional and Post-Conflict Societies*. Ed. by Julia Viebach and Christiane Ulbrich. London: Routledge, pp. 89–108.

Graham, Stephen (2004). “Postmortem City: Toward an Urban Geopolitics”. In: *Cities, War, and Terrorism: Towards an Urban Geopolitics*. Ed. by Stephen Graham. Oxford: Blackwell, pp. 1–30.

— (2006). “Cities, War, and Terrorism: Towards an Urban Geopolitics”. In: *International Journal of Urban and Regional Research* 30.1, pp. 1–42.

— (2008). “Robowar Dreams: US Military Technophilia and Global South Urbanisation”. In: *City* 12.1, pp. 25–49.

— (2011). *Cities Under Siege: The New Military Urbanism*. London: Verso.

— (2012). “When Life Itself Is War: On the Urbanization of Military and Security Doctrine”. In: *International Journal of Urban and Regional Research* 36.1, pp. 136–155.

Gurney, Craig M. (2023). “Dangerous liaisons? Applying the social harm perspective to the social inequality, housing and health trifecta during the COVID-19 pandemic”. In: *International Journal of Housing Policy* 23.2, pp. 232–259.

Hälterlein, Jens (2021). “Epistemologies of Predictive Policing: Mathematical Social Science, Big Data, and the Rearticulation of Pattern”. In: *Big Data & Society* 8.1.

Hanafi, Sari (2012). “Explaining Spacio-cide in the Palestinian Territory: Colonization, Separation, and the Logic of Elimination”. In: *Current Sociology* 61.2, pp. 190–211.

Haraway, Donna (1988). “Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective”. In: *Feminist Studies* 14.3, pp. 575–599.

Hogue, Simon (2021). “Project Maven, Big Data, and Ubiquitous Knowledge: The Impossible Promises and Hidden Politics of Algorithmic Security Vision”. In: *Automating Crime Prevention, Surveillance, and Military Operations*. Ed. by Aleš Završnik and Vasja Badalič. Cham: Springer.

Holail, Tamer, Mohamed Abou Donia, and Maher El-Hallaq (2024). “Time-Series Satellite Remote Sensing Reveals Gradually Increasing War Damage in the Gaza Strip”. In: *National Science Review* 11, nwae304.

Hourani, Sara (2024). “In Dangerous Times: War, Reconstruction and the City”. In: *City* 28.1–2, pp. 1–26.

Irungbam, Robinson S. (2026). “Repressive Connectivity and Digital Resistance: Internet Shutdowns as Epistemicide in Manipur”. In: *Information, Communication & Society*. OnlineFirst.

Jacobs, Niels and Mark Van Ostaijen (2024). “Smart Cities as “Big Brother Only to the Dialogue Masses”: The Eindhoven Nightlife Living Lab”. In: *Urban Studies* 61.3, pp. 495–513.

Johnsen, Frank T., Zbigniew Zielinski, Konrad Wrona, Niranjan Suri, Christoph Fuchs, and Manas Pradhan (2018). “Application of IoT in Military Operations in a Smart City”. In: *2018 International Conference on Military Communications and Information Systems (ICMCIS)*. IEEE.

Kaufmann, Mareile (2019). “Predictive Policing and the Politics of Patterns”. In: *The British Journal of Criminology* 59.3, pp. 674–692.

King, Peter (2004). *Private Dwelling*. London: Routledge.

Kitchen, Veronica and Kim Rygiel (2014). “Privatizing Security, Securitizing Policing: The Case of the G20 in Toronto, Canada”. In: *International Political Sociology* 8.2, pp. 201–217.

Kitchin, Rob (2014). “Big Data, New Epistemologies and Paradigm Shifts”. In: *Big Data & Society* 1.1, p. 2053951714528481.

— (2018). “Reframing, Reimagining and Remaking Smart Cities”. In: *Programmable City Working Paper* 37.

Lotfi-Jam, Farzin (2025). “Realtime Urbanism: The Architecture of Packets, Pixels, and Neurons”. In: *International Journal of Urban and Regional Research*. OnlineFirst, pp. 158–183.

Lyon, David (2014). “Surveillance and the Eye of God”. In: *Studies in Christian Ethics* 27.1, pp. 21–32.

Marvin, Simon and Andrés Luque-Ayala (2017). “Urban Operating Systems: Diagramming the City”. In: *International Journal of Urban and Regional Research* 41.1, pp. 84–103.

Massaro, Vanessa A. (2015). “The Pacification of the American City: The MOVE Bombing and the Militarization of American Urban Policing”. In: *Urban Geography* 36.8, pp. 1138–1152.

Meade, Taryn (2011). “Violence and Domestic Space: Demolition as a Means of Control in the West Bank”. In: *Cultural Geographies* 18.4, pp. 433–458.

Michael, Kobi (2007). “The Israel Defense Forces as an Epistemic Authority: An Intellectual Challenge in the Reality of the Israeli–Palestinian Conflict”. In: *Journal of Strategic Studies* 30.3, pp. 421–446.

Miller, Peter and Nikolas Rose (1990). “Governing Economic Life”. In: *Economy and Society* 19.1, pp. 1–31.

Page, Matthew J., Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, et al. (2021). “The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews”. In: *BMJ* 372, n71.

Pasquale, Frank (2015). *The Black Box Society: The Secret Algorithms That Control Money and Information*. Cambridge, MA: Harvard University Press.

Paz-Fuchs, Amir (2010). “The Social Contract Revisited: The Purpose of Home Demolitions Under International Law”. In: *European Journal of International Law* 21.1, pp. 165–187.

Peteet, Julie (2015). “Camps and Enclaves: Palestine in the Time of Closure”. In: *Journal of Palestine Studies* 45.1, pp. 30–51.

Pradhan, Manas (2018). “A Survey of Smart City Assets for Future Military Usage”. In: *2018 International Symposium on Networks, Computers and Communications (ISNCC)*. IEEE.

Priem, Jason, Heather Piwowar, and Richard Orr (2022). “OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts”. In: *arXiv preprint arXiv:2205.01833*.

Rechtbank Den Haag (2020). *NJCM c.s. vs De Staat der Nederlanden (SyRI), ECLI:NL:RBDHA:2020:865*. District Court of The Hague, 5 February 2020.

Riberto, Giulio, Marco Govoni, Cesare Stefanelli, Niranjan Suri, and Mauro Tortonesi (2018). "Leveraging Civilian IoT Infrastructures to Support Warfighting Activities in Urban Environments". In: *2018 IEEE 4th World Forum on Internet of Things (WF-IoT)*. IEEE, pp. 118–123.

Robinson, Nick (2020). "China's Smart Cities as Geopolitical Instruments". In: *The RUSI Journal* 165.5–6, pp. 56–67.

Royal Commission into the Robodebt Scheme (2023). *Final Report*. Commonwealth of Australia. Canberra.

Sadowski, Jathan (2026). "Machine's Eye View: Postmodern Data Science and the Politics of Ground Truth". In: *Science, Technology, & Human Values* 51.2, pp. 251–276.

Salamanca, Omar J. (2011). "Unplug and Play: Manufacturing Collapse in Gaza". In: *Accumulation by Dispossession: Transformative Cities in the New Global Order*. Ed. by Swapna Banerjee-Guha. New Delhi: SAGE, pp. 59–82.

Samara, Tony Roshan (2010). "Policing Development: Urban Renewal as Neo-liberal Security Strategy". In: *Urban Studies* 47.1, pp. 197–214.

Scannell, R. J. (2019). "This Is Not Minority Report: Predictive Policing and Population Racism". In: *Captivating Technology: Race, Carceral Technoscience, and Liberatory Imagination in Everyday Life*. Ed. by Ruha Benjamin. Durham, NC: Duke University Press, pp. 107–129.

Schimmel, Noam (2021). "The NFL as a Pentagon Supplier: Fans as Citizen-Soldiers". In: *Journal of Sport and Social Issues* 45.4, pp. 335–357.

Schoot, Rens van de, Jonathan de Bruin, Raoul Schram, Parisa Zahedi, Jan de Boer, Felix Weijdema, Bianca Kramer, Martijn Huijts, Maarten Hoogerwerf, Gerbrich Ferdinands, Albert Harkema, Joukje Willemsen, Yongchao Ma, Qixiang Fang, Sybren Hindriks, Lars Tummers, and Daniel L. Oberski (2021). "An open source machine learning framework for efficient and transparent systematic reviews". In: *Nature Machine Intelligence* 3, pp. 125–133.

Schuilenburg, Marc and Rik Peeters (2018). “Smart Cities and the Architecture of Security: Pastoral Power and the Scripted Design of Public Space”. In: *City, Territory and Architecture* 5.1, pp. 1–11.

Shalhoub-Kevorkian, Nadera (2017). “The Occupation of the Senses: The Prosthetic and Aesthetic of State Terror”. In: *The British Journal of Criminology* 57.6, pp. 1272–1291.

Sharma, Sarah and Jasbinder Nijjar (2024). “Algorithmic Patrol: The Futures of Predictive Policing”. In: *Science, Technology, & Human Values* 49.2, pp. 387–412.

Smith, Chris (2022). “Leviathan’s Architect: Urbicide, Urban Renewal, and the Right to the City in Turkey’s Kurdish Southeast”. In: *City* 26.4, pp. 613–636.

Srivastava, Sanjay (2021). “Algorithmic Governance and the International Politics of Big Tech”. In: *Perspectives on Politics* 19.4, pp. 1125–1141.

Suri, Niranjan, Zbigniew Zielinski, Mauro Tortonesi, Christoph Fuchs, Manas Pradhan, Konrad Wrona, Janusz Furtak, and Dragos Bogdan Vasilache (2018). “Exploiting Smart City IoT for Disaster Recovery Operations”. In: *2018 IEEE 4th World Forum on Internet of Things (WF-IoT)*. IEEE.

Sutton, Anthea, Mark Clowes, Louise Preston, and Andrew Booth (2019). “Meeting the review family: exploring review types and associated information retrieval requirements”. In: *Health Information & Libraries Journal* 36.3, pp. 202–222.

Tawil-Souri, Helga (2011). “Qalandia Checkpoint as Space and Nonplace”. In: *Space and Culture* 14.1, pp. 4–26.

Tenney, Matthew (2016). “Algorithmic Regulation and the Limits of the Digital City”. In: *Urban Geography* 37.5, pp. 729–745.

Townsend, Anthony M. (2013). *Smart Cities: Big Data, Civic Hackers and the Quest for a New Utopia*. New York: W.W. Norton & Company.

Tranfield, David, David Denyer, and Pam Smart (2003). “Towards a Methodology for Developing Evidence-Informed Management Knowledge by Means of Systematic Review”. In: *British Journal of Management* 14, pp. 207–222.

Tsiolis, Konstantinos and Aris Leivadeas (2025). “A Smart City Assisted Internet of Military Defense Things Architecture”. In: *IEEE Internet of Things Journal*. OnlineFirst.

Tyndall, Jessica (2010). *AACODS Checklist*. Flinders University, Adelaide.

UN Independent International Commission of Inquiry on Ukraine (2025). *Drone attacks targeting civilians in the Kherson region*. `https://www.ohchr.org/en/hr-bodies/hrc/iicihr-ukraine/index`. Report characterising short-range drone attacks on civilians as crimes against humanity.

Volinz, Lior (2019). "From Above and Below: Surveillance, Religion and the Politics of Low-Altitude Security". In: *Security Dialogue* 50.4, pp. 343–359.

Wall, Tyler (2011). "Vertical Security and the Megacity". In: *Theory & Event* 14.2.

— (2013). "Unmanning the Police Manhunt: Vertical Security as Pacification". In: *Socialist Studies/Études Socialistes* 9.2, pp. 32–56.

Weber, Valentin (2023). "The Chinese City Brain and the Future of Urban AI Governance". In: *Big Data & Society* 10.2.

Weizman, Eyal (2007). *Hollow Land: Israel's Architecture of Occupation*. London: Verso.

— (2017). *Forensic Architecture: Violence at the Threshold of Detectability*. New York: Zone Books.

Wiig, Alan (2018). "Secure the City, Revitalize the Economy: Smart Cities and the Militarization of Urban Government". In: *Urban Geography* 39.7, pp. 1035–1052.

Wilhelmsen, Eira (2022). "Commercial Data, National Security: Location Intelligence and the Cellebrite Controversy". In: *Surveillance & Society* 20.3, pp. 270–284.

Williamson, Ben (2015). "Programming Environments: Environmentality and Citizen Sensing in the Smart City". In: *Cultural Geographies* 22.1, pp. 173–191.

Wilson, David and Elvin Wyly (2026). "Dracula Urbanism and Smart City Mania: Urban Change in the Twilight of Neoliberalism". In: *Urban Geography* 47.1, pp. 1–40.

Wohlin, Claes (2014). "Guidelines for Snowballing in Systematic Literature Studies and a Replication in Software Engineering". In: *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE '14)*. ACM, pp. 1–10.

Završnik, Aleš and Vasja Badalič, eds. (2021). *Automating Crime Prevention, Surveillance, and Military Operations*. Cham: Springer.

Zeitoun, Mark (2018). “Urban Warfare Ecology: A Study of Water Supply in Basrah”. In: *International Review of the Red Cross* 99.906, pp. 1213–1238.

Ziadah, Rafeef (2021). “Surveillance, Race, and Social Sorting in the United Arab Emirates”. In: *Surveillance & Society* 19.4, pp. 413–428.

Appendix A

AI-Use Statement

This dissertation was produced with AI assistance under the University of Glasgow's guidelines on the use of generative AI in assessment. A large-language-model classifier (DeepSeek V4 Flash; temperature 0.0) was used as a screening and coding assistant during the systematic literature review: the model assigned provisional include/exclude verdicts and theme scores against a written codebook, with all outputs reviewed, verified through random sampling, and final decisions retained by the author. Drafting assistance was used for initial chapter scaffolding and prose generation; all agent-generated text was reviewed, revised, and substantially edited by me. The AI-use workflow is fully documented in the SLR log and analysis directories accompanying this submission, publicly archived at `https://github.com/EVasiloudes/MSc-PPM-Dissertation---Repo`. No AI-generated text was incorporated without authorial review. The workflow was discussed with the dissertation supervisor, Craig Gurney, who confirmed that its use, as disclosed, is in compliance with the applicable University guidance.

Appendix B

Systematic Search Strategy

This appendix does not count toward the word limit.

The final search strings, executed on 3 July 2026, are reproduced verbatim below. Database-specific syntax and limiters are preserved; record counts are reported in the PRISMA flow diagram (Figure 3.3).

B.1 Search A — Civil-Military Urban Technology Nexus

B.1.1 Web of Science (Core Collection)

```
TS=("algorithmic governance" OR "urban analytics" OR "smart cit*" OR
"platform urbanism" OR "predictive policing" OR "urban big data" OR
"data-driven urbanism") AND TS=("dual-use" OR "civil-military" OR military
OR militari* OR "national security" OR securitis* OR securitiz* OR weaponi*
OR warfare OR "state surveillance" OR "mass surveillance" OR OSINT) AND
PY=(1990-2025)
```

B.1.2 Scopus

```
TITLE-ABS-KEY(( "algorithmic governance" OR "urban analytics" OR "smart
cit*" OR "platform urbanism" OR "predictive policing" OR "urban big data" OR
"data-driven urbanism" ) AND ( "dual-use" OR "civil-military" OR "military"
OR militari* OR "national security" OR securitis* OR securitiz* OR weaponi*
```

```
OR warfare OR "state surveillance" OR "mass surveillance" OR "OSINT" )) AND
PUBYEAR > 1989 AND PUBYEAR < 2026 AND ( LIMIT-TO ( SUBJAREA,"DECI" ) OR
LIMIT-TO ( SUBJAREA,"SOCI" ) OR LIMIT-TO ( SUBJAREA,"BUSI" ) OR LIMIT-TO (
SUBJAREA,"ECON" ) OR LIMIT-TO ( SUBJAREA,"ARTS" ) OR LIMIT-TO (
SUBJAREA,"MULT" ) ) AND ( LIMIT-TO ( DOCTYPE,"ar" ) OR LIMIT-TO (
DOCTYPE,"ch" ) OR LIMIT-TO ( DOCTYPE,"re" ) ) AND ( LIMIT-TO (
LANGUAGE,"English" ) )
```

B.1.3 Policy Commons

```
summary:("algorithmic governance" OR "urban analytics" OR "smart city" OR
"smart cities" OR "platform urbanism" OR "predictive policing" OR "urban big
data" OR "data-driven urbanism") AND summary:("dual-use" OR "civil-military"
OR military OR "national security" OR warfare OR securitisation OR
securitization OR weaponisation OR weaponization OR "state surveillance" OR
"mass surveillance")
```

B.2 Search B — Spatial Destruction and the Datafied City

B.2.1 Web of Science (Core Collection)

```
TS=("domicide" OR "urbicide" OR "spatial violence" OR "home unmaking" OR
"urban destruction" OR "urban warfare") AND TS=(data OR algorithm OR AI OR
surveillance OR GIS OR "remote sensing") AND PY=(1990-2025)
```

B.2.2 Scopus

```
TITLE-ABS-KEY(("domicide" OR "urbicide" OR "spatial violence" OR "home
unmaking" OR "urban destruction" OR "urban warfare") AND ("data" OR
"algorithm" OR "AI" OR "surveillance" OR "GIS" OR "remote sensing")) AND
PUBYEAR > 1989 AND PUBYEAR < 2026 AND ( LIMIT-TO ( SUBJAREA,"DECI" ) OR
LIMIT-TO ( SUBJAREA,"SOCI" ) OR LIMIT-TO ( SUBJAREA,"BUSI" ) OR LIMIT-TO (
SUBJAREA,"ECON" ) OR LIMIT-TO ( SUBJAREA,"ARTS" ) OR LIMIT-TO (
```

```
SUBJAREA,"MULT" ) ) AND ( LIMIT-TO ( DOCTYPE,"ar" ) OR LIMIT-TO (
DOCTYPE,"ch" ) OR LIMIT-TO ( DOCTYPE,"re" ) ) AND ( LIMIT-TO (
LANGUAGE,"English" ) )
```

B.2.3 Policy Commons

```
title:("domicide" OR "urbicide" OR "spatial violence" OR "home unmaking" OR
"urban destruction" OR "urban warfare") AND summary:(data OR algorithm OR AI
OR surveillance OR GIS OR "remote sensing")
```

B.3 Notes

Search fields: Web of Science Topic (title, abstract, author keywords, Keywords Plus); Scopus TITLE-ABS-KEY (title, abstract, keywords); Policy Commons title and summary fields. Scopus searches were limited to decision sciences, social sciences, business, economics, arts and humanities, and multidisciplinary subject areas; to articles, book chapters, and reviews; and to English-language records. Search strings were calibrated through the scoping iterations described in §3.3, and the disclosed deviations from the formative protocol (including the Overton substitution) are documented in §3.3.1 and §3.6.